

Dendritic morphology as a dictionary for local credit assignment

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Abstract

Dendritic trees compartmentalize synaptic inputs, but which spatial distinctions a learning signal must preserve is unclear. We describe delivered credit as combinations of spatial profiles constrained by morphology. In image classifiers with one dendritic neuronal layer, preserving neuronal identity accounts for most of the feedback benefit. Contextual conflict favors branch selection, while hierarchical distractors favor ancestry grouping at a bandwidth predicted by the task. Targets with matched input-sensitivity spectra develop different learned credit spectra and tolerance of fixed profiles; a local inhibitory gate also supports learning of opposing branch tuning in a conductance tree. Serial compartments help when nuisance-factor information is distributed across modules. Reconstructed arbors compress modeled passive perturbation fields through ancestry profiles, and focal shunts regulate their gains with selectivity set by electrotonic state. Measured visual responses provide no evidence of preferential ancestry alignment in the sampled cells. The framework links task structure to useful credit distinctions while separating available delivery patterns from the mechanisms that generate their coefficients.

Introduction

Learning must link a synapse’s local activity to its contribution to a circuit outcome. Cable filtering, nonlinear integration and spatially organized inhibition make locations within a dendritic tree functionally distinct [1–7]. These differences expand computation. We ask when branching also helps distribute useful learning signals.

Credit assignment asks how changing a synapse would affect task error. Backpropagation (BP) answers this by calculating the loss gradient—the sensitivity of task error to each parameter [8]; related sensitivity calculations apply to recurrent and steady-state networks [9, 10]. For a point neuron, each weight gradient combines a local sensitivity, called eligibility, with a neuron-specific learning signal. Hebbian plasticity captures the association between local pre- and postsynaptic variables [11], while three-factor rules combine that eligibility with an outcome-dependent signal [12]. We call a rule local when a synapse combines its own eligibility with an available learning signal, whether or not that signal is computed locally.

Biologically motivated learning theories differ in how they obtain such signals. Feedback alignment communicates errors through separate pathways [13, 14]; equilibrium, predictive and prospective methods infer target states or errors from dynamics [15–17]; eligibility propagation combines temporal traces with neuron-specific signals [18]. One number shared across neurons carries less task information than separately assigned neuronal signals, although each synapse still has its own eligibility [19–21]. Recent causal evidence for distinct instructive signals across neuronal populations motivates asking what additional distinctions are useful within a neuron’s arbor [22].

A *route dictionary* describes the spatial profiles available for delivering learning signals. A route identifies the locations reached; its profile specifies their relative amplitudes, and a trial-dependent coefficient scales that profile. Each dictionary column stores one profile. A shared profile broadcasts throughout an arbor; combining several profiles can distinguish subtrees. We use *basis* for linearly independent profiles and *dictionary* when profiles may be redundant. Morphology constrains the available routes; the task and teaching mechanism determine useful coefficients.

Conductance changes the gains of anatomical delivery routes. Excitatory and inhibitory contacts open nonnegative conductances whose reversal potentials determine current; inhibition can lower input resistance and attenuate other

inputs even when its own current is small [23–27]. Rescaling a delivery channel preserves its spatial profile, whereas altered cable transport can reshape how local voltage affects somatic output and task error.

Dendritic models already connect nonlinear compartments and structural plasticity to memory capacity [28], and branch-specific plasticity can organize nonlinear feature combinations [29]. Successive computations in artificial trees and detailed cortical models also establish substantial forward capacity [30, 31]. Dendritic learning theories supply mechanisms for somatic prediction, segregated teaching, interneuron-mediated errors, bursts and prospective dynamics [32–38]; cable-learning models optimize reconstructed neurons using synaptic sensitivities [39]. We ask which spatial distinctions these mechanisms must preserve within an arbor.

The conductance-tree gradient factorization and LocalCA rule were introduced in our predecessor preprint [40]. This Article extends those foundations with controlled task-to-credit comparisons, matched-spectrum and conductance learning experiments, and anatomical transport and response tests.

We test a progression from neuronal identity to branch selection, ancestry grouping and example-dependent credit across interacting branches. Controlled learning tasks establish useful distinctions; forward-compatibility tests separate learning gaps from representational limits. Reconstructed arbors then test anatomical delivery capacity and conductance-dependent gains, while measured responses bound the biological interpretation.

Results

Exact gradients separate local sensitivity from downstream credit

For a point neuron $y_u = f_u(\sum_i w_i x_i + b_u)$, the chain rule gives $\partial \mathcal{L} / \partial w_i = x_i f'_u \delta_u$, where x_i and w_i are synaptic input and weight, b_u is bias, and $\delta_u = \partial \mathcal{L} / \partial y_u$ is the neuronal error. The derivative f'_u is evaluated at the current input sum. Local eligibility $x_i f'_u$ and task-derived credit therefore supply different information (Fig. 1A).

The rooted-tree model has soma $n = 0$, compartment voltage V_n , and nonnegative synaptic conductances g_i . A contact with reversal potential E_i^{rev} and nonnegative activity x_i contributes current $g_i x_i (E_i^{\text{rev}} - V_n)$. In the directed artificial trees, child outputs drive their parents and a terminal-to-soma pass computes the steady state. Total input resistance R_n^{tot} is the reciprocal of the summed leak, excitatory, inhibitory and child-coupling conductances (Methods). Differentiating the current balance yields

$$\frac{\partial \mathcal{L}}{\partial g_i} = \underbrace{x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n)}_{\text{local eligibility}} \underbrace{\delta_{n,u}^V}_{\partial \mathcal{L} / \partial V_n}. \quad (1)$$

Excitatory (E) and inhibitory (I) contacts share this factorization (Fig. 1A). Eligibility tells how conductance changes local voltage; compartment credit tells how that voltage changes task loss.

Voltage error denotes a loss derivative. Each compartment has one directed path to the soma, giving

$$\delta_{n,u}^V = \delta_{0,u}^V \tilde{\alpha}_n, \quad \tilde{\alpha}_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightarrow 0)} f'_i(V_i) R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}. \quad (2)$$

Here $f'_i(V_i)$ is the child compartment’s output derivative, $g_{i \rightarrow k}^{\text{den}}$ is its coupling conductance to parent k , and $\delta_{0,u}^V = f'_0(V_0) \delta_u$ is the somatic voltage error. The path multiplier $\tilde{\alpha}_n$ converts one somatic error into a spatial field. The conductance-only component of this product is attenuated by additional conductance on a path’s ancestors; branch nonlinearities and forward voltages can change as well (Supplementary Section S1).

Reciprocal cables instead require a joint adjoint calculation of how local currents affect task loss (Methods). These mathematical sensitivities do not establish a biological error-transmission mechanism.

Local credit assignment (LocalCA) retains eligibility and supplies an approximate compartment signal, denoted by a hat:

$$\hat{V}_{g_i} \mathcal{L} = x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n) \hat{\delta}_{n,u}^V, \quad (3)$$

$$\hat{\delta}_u^V = A_u \mathbf{c}_u. \quad (4)$$

Columns of $A_u \in \mathbb{R}^{N_u \times K}$ are fixed spatial delivery profiles over N_u routed nonsomatic compartments; $\mathbf{c}_u$ contains their example-dependent coefficients. To distinguish coefficient generation from delivery gain, we can write $\mathbf{c}_u = \Gamma_u \mathbf{c}_u^{\text{source}}$, where diagonal Γ_u scales each profile and $\mathbf{c}_u^{\text{source}}$ is the supplied signal (Fig. 1A). Broadcast, subtree and compartment profiles give increasingly detailed spatial addresses (Fig. 1B). This is a delivery model, distinct from the forward cable equations and the mechanism estimating coefficients.

At $K = 1$, the all-ones column $A_u = \mathbf{1}$ broadcasts one coefficient through the arbor; at $K = N_u$, the identity matrix $A_u = I$ can reproduce the full nonsomatic field. Nested subtrees provide intermediate dictionaries. Profile count, rank (the number of independent profiles) and coefficient information are distinct resources. Exact path transport combines one somatic scalar with state-dependent local derivatives; it does not require an independent external error for every compartment.

Capture is the fraction of squared learning-signal magnitude that a dictionary can reproduce. For a nonzero field $\boldsymbol{\delta}$,

$$\mathcal{C}(A; \boldsymbol{\delta}) = \frac{\|P_A \boldsymbol{\delta}\|^2}{\|\boldsymbol{\delta}\|^2}, \quad P_A = AA^\dagger.$$

Here P_A gives the least-squares reconstruction, $A^\dagger$ is the Moore–Penrose pseudoinverse and $\|\cdot\|$ the Euclidean norm. Capture measures field energy rather than decision accuracy. Averaging these per-field ratios weights fields equally; $\text{tr}(P_A C) / \text{tr}(C)$, with $C = \mathbb{E}[\boldsymbol{\delta} \boldsymbol{\delta}^\top]$, is the fraction of total ensemble energy captured, weighting fields by their energy. The uncentered second-moment matrix C has trace $\text{tr}(C)$, the sum of its diagonal entries. These capture measures can differ, and both assume optimal projection coefficients rather than an implemented encoder.

The field coordinate also matters. In nonlinear image models, we route errors of compartment outputs and retain the local output derivative in eligibility; voltage-error capture is reported separately. Identity-output conductance trees use voltage errors directly. Supplementary Table S2 compares the coordinates, coefficient sources and state dependence. Learning experiments test whether this available capacity improves performance.

Projection can also filter update noise outside a dictionary’s span. Retaining most of the useful gradient while admitting less noise can improve a one-step guarantee (Fig. 1C). If the span contains the entire gradient, its mean is preserved. A one-step smoothness bound formalizes this balance between retained descent and update variance (Methods; Supplementary Fig. S2). Its initialization utility did not select the best final morphology: in a separate 12,800-fit prospective experiment, it lost to a fixed maximum-budget tree in all twenty seed averages in both selection arms (Supplementary Fig. S33). We therefore use the bound to interpret local filtering, not to select a final tree.

One signal per neuron captures most of the image-task learning benefit

Image classification first tests whether within-arbor resolution helps when neuronal errors are available. All image networks used one dendritic layer and a trainable readout, which supplied exact neuronal output errors before the primary feedback interventions. The MNIST comparison used 128 directed $[3, 3]$ trees, each with three proximal and nine terminal compartments, receiving flattened images. Shunting and raw-additive architectures shared the graph, initial contact construction, nonlinearities and training protocol, with architecture-specific initialization. The additive control removed reversal-potential driving forces and the conductance denominator, retaining signed current summation without shunting.

Six rules separated neuronal identity, spatial resolution and learning in the dendritic core (Fig. 1B,D). A strict scalar shared one signal across neurons; neuron-specific feedback broadcast each neuron’s own output error through its arbor. A projected one-profile rule averaged the exact activation errors over all twelve nonsomatic compartments, while three subtree profiles averaged them within each proximal compartment and its three children. Both used the same exact field and preserved its mean at identical parameters, while allowing different norms. Exact transport retained every compartment error. A decoder-only reference trained the readout while freezing the core. Three development seeds selected each rule’s rate from three common multipliers; ten fresh paired seeds per architecture evaluated selected and original common rates over 180 epochs (Methods; Supplementary Fig. S7A,B).

At selected rates, neuron-specific feedback exceeded the strict scalar by 9.23 percentage points in shunting trees and 7.46 points in additive trees, reaching 97.12% and 97.11% accuracy. Decoder-only learning reached 80.96% and 89.57%, respectively (Fig. 1D). Increasing the projected dictionary from one to three profiles changed accuracy by

only 0.049 points in shunting trees (95% paired interval, -0.048 to 0.141) and 0.036 points in additive trees (-0.056 to 0.122 ; Fig. 1E). The common-rate comparison gave similarly small differences (Supplementary Fig. S7C). Exact transport added 0.010 points over neuron-specific feedback in shunting trees, but 0.182 points in additive trees (0.114 – 0.252 ; positive in all ten seeds). Both exact transport and one projected profile reached about 97.3% in additive trees. Most learning benefit therefore came from preserving neuronal coordinates; the smaller additive improvement was attainable without separating subtrees.

The richer dictionary nevertheless captured more of the trained activation-error field in the same cohort. Moving from one to three profiles raised mean capture from 0.622 to 0.655 in shunting trees and from 0.764 to 0.859 in additive trees (Fig. 1F). These ratios average over example–neuron fields and use the coordinates delivered during learning. Thus improved reconstruction did not yield a comparable accuracy benefit. Earlier voltage-credit diagnostics and recipient-permutation controls support spatial field structure and the importance of neuronal identity, respectively (Supplementary Section S2 and Fig. S7D–G).

With fixed random soma-level feedback, neuron-specific signals improved over scalar feedback by about eleven points, while exact paths added at most 0.09 points (fifteen paired seeds; Fig. 1G; Supplementary Fig. S6D,E). This control changes weight transport but retains the single dendritic layer and the same output-error dimensionality. Fashion-MNIST also favored neuronal feedback. On flattened CIFAR-10, exact paths reduced accuracy relative to neuronal feedback by 0.86 points (0.55 – 1.17), while remaining equivalent to matched backpropagation within the predefined margin (Fig. 1G; Supplementary Fig. S6A–C).

Thus finer delivery need not improve learning even when the error field has spatial structure. Further depth and task-generator controls are in Supplementary Figs. S6 and S8. We next introduce tasks that make specific spatial distinctions useful.

Branch-selective information prevents updates from inactive distractors

Branch selection is useful when otherwise valid local activity comes from irrelevant inputs. We tested this with one logistic output and $B \in \{2, 4, 8\}$ parallel branch weight vectors and no internal forward compartments or conductance nonlinearity (Fig. 2A). Context selected the branch determining the output, while every branch received an image and formed an ungated local eligibility.

Inputs were average-pooled, standardized Fashion-MNIST images from classes 0 and 6. The selected image determined the binary target. At conflict dose $\chi = 0$, all branches received that image; increasing χ independently replaced nonselected inputs with opposite-class images. For example i , let b_i^* be the selected branch, $\mathbf{x}_{i,b}$ its input vector and $y_i \in \{0, 1\}$ the target. The branch weights $\mathbf{w}_b$ give the pre-probability score (logit) and output error

$$z_i = \mathbf{w}_{b_i^*}^T \mathbf{x}_{i,b_i^*}, \quad \delta_i = \sigma(z_i) - y_i. \quad (5)$$

Here $\sigma(z) = 1/(1 + e^{-z})$ is the logistic function. Unselected branches have zero exact derivative despite nonzero eligibility. Conflict therefore challenges a rule that ignores selection by exposing it to irrelevant, opposite-class eligibilities.

For N examples, correctly selected and neuron-shared gradient estimates were

$$\mathbf{d}_b^{\text{selected}} = N^{-1} \sum_i \delta_i \mathbf{1}[b_i^* = b] \mathbf{x}_{i,b}, \quad \mathbf{d}_b^{\text{shared}} = N^{-1} \sum_i \delta_i B^{-1} \mathbf{x}_{i,b}. \quad (6)$$

The indicator $\mathbf{1}[\cdot]$ is one when its condition holds, zero otherwise; $1/B$ matches scale without supplying branch information. A feedback selector and a branch-local eligibility gate give the same update (Fig. 2B), testing access to selection rather than a need for independently computed external errors.

With centered, symmetric class evidence, the shared-mode prediction crosses zero at $\chi_c = B/[2(B-1)]$: 1, 2/3 and 4/7 for two, four and eight branches (Fig. 2C; Methods). More branches predict earlier loss of compatibility; this mean-field boundary need not equal a trained chance crossing.

At zero conflict, selected and shared rules achieved similar accuracy. Their difference increased with conflict, reaching 35–58 percentage points at full conflict across the tested branch counts (20 paired seeds; Fig. 2D). In every seed, the advantage of correct routing increased with conflict. A cyclic reassignment sent every signal to the wrong

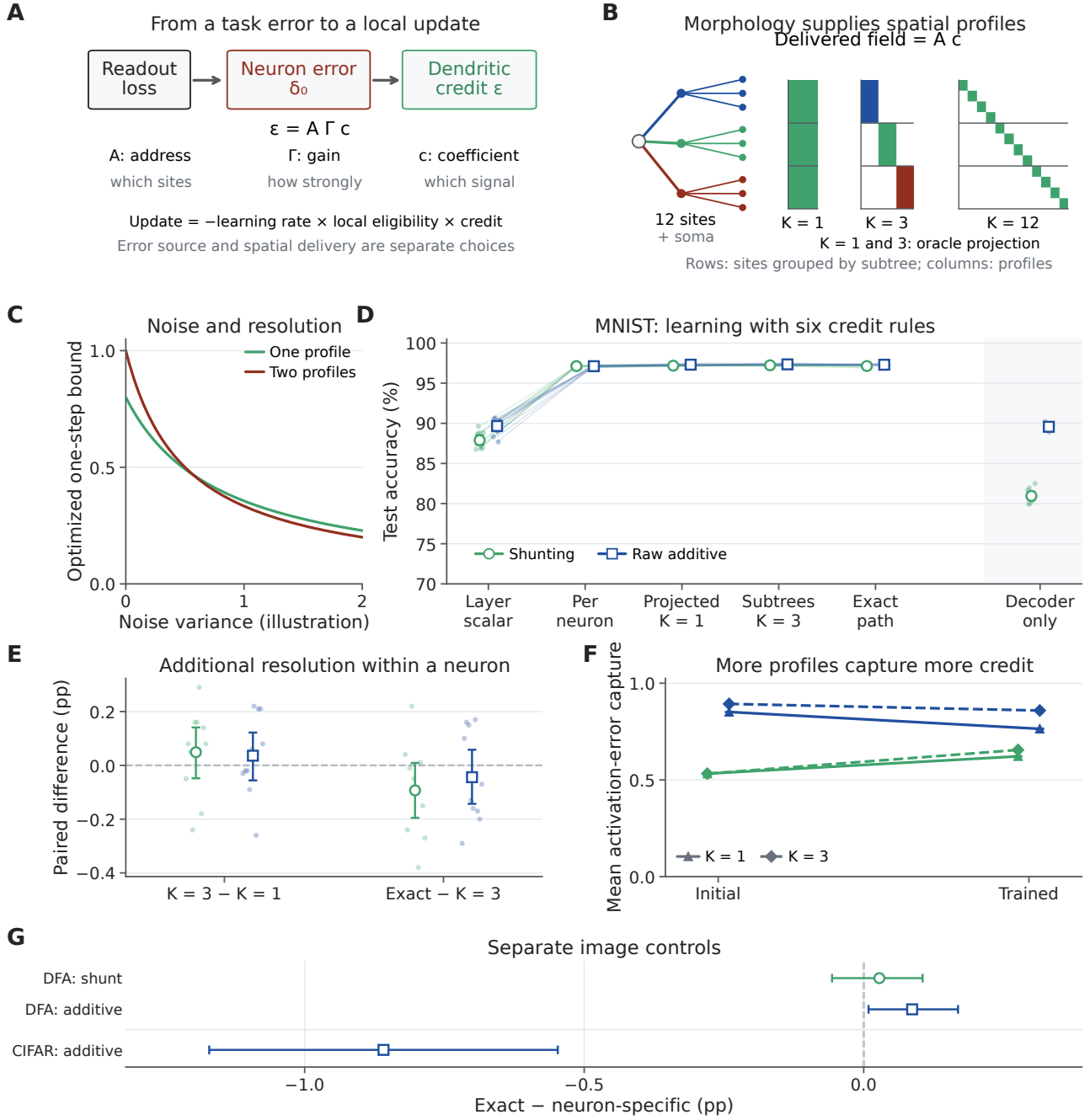


Figure 1: Task-derived credit separates neuronal identity, spatial address and gain. **A**, Neuron-specific coefficients c , dictionary A and diagonal gains Γ give credit $\epsilon = A\Gamma c$. Updates multiply credit by local eligibility and negative learning rate. **B**, Twelve nonsomatic sites with $K = 1$ common, $K = 3$ subtree or $K = 12$ site profiles. Projected rules use oracle activation-error coefficients and retain exact somatic errors. **C**, Illustrative projection of a unit-energy gradient under isotropic noise variance σ^2 . A rank- K span retains energy q . Twice the smoothness constant L times the optimized one-step bound is $q^2/(q + K\sigma^2)$, comparing $(K, q) = (1, 0.8)$ and $(2, 1)$. This local bound does not predict final structure. **D**, MNIST with one dendritic layer and a linear readout. Per-neuron credit broadcasts exact somatic activation errors; decoder-only freezes the core. Lines pair ten fresh seeds per architecture. Three development seeds selected rates; validation selected checkpoints within 180 epochs. **E**, Paired effects of additional spatial resolution. **F**, Initial and trained capture in the same exact-rule seeds averages squared-energy ratios over nonzero fields from 2,048 images and 128 neurons. Solid triangles and dashed diamonds denote $K = 1$ and $K = 3$; architecture colors match **D**. **G**, Separate MNIST direct-feedback-alignment (DFA; fifteen seeds) and flattened CIFAR-10 (twenty seeds) controls. Symbols and intervals in **D–G** show means and descriptive 95% seed-bootstrap intervals. pp, percentage points.

branch at matched rank and sparsity. It was harmful even at zero conflict, reducing accuracy by 12–25 points relative to correct selection. Analytic backpropagation and an explicitly gated grouped-point model matched correct selection exactly, as required by their algebraic equivalence (Fig. 2E; Supplementary Section S3).

The shared rule reached chance at lower conflict as branch count increased, consistent with the predicted shift (Fig. 2F); five two-branch seeds never crossed in the sampled range. Gradient alignment also changed during learning (Methods; Supplementary Fig. S9C–E). Credit reaching inactive branches can therefore harm learning. A separate two-stream interference task tests this principle (Supplementary Fig. S9A,B).

Ancestry provides a modest advantage at an intermediate feedback budget

Between shared and branch-resolved feedback, nested subtree routes offer an intermediate dictionary. The task contained eight independent, eight-dimensional terminal weight blocks at the leaves of a balanced binary feedback tree. Only the context-selected block determined the logit. Internal tree nodes defined feedback supports and did not perform serial forward computation (Fig. 3A).

Distractors opposed the selected stream’s class signal increasingly with distance through the feedback hierarchy, deliberately favoring task-aligned grouping. At each budget $K \in \{1, 2, 4, 8\}$, the routing field had K independent patterns and unit-length rows. Ancestry routes grouped the terminals into the whole tree, halves, sibling pairs or individual terminals. Permuting task streams across leaves changed alignment without changing the streams or forward parameter budget. Here bandwidth counts independent feedback patterns, not physical stages or independent scalar losses; other single-pattern controls need not be uniform. The complete generator and update are given in Methods.

The generator predicts the budget at which ancestry can help. The selected stream has class coefficient one; its sibling, same-half nonsiblings and other-half streams have coefficients -0.15 , -0.45 and -0.75 . The unnormalized sum of the class signal within the selected ancestry group gives -3.05 , -0.05 , 0.85 and 1 at $K = 1, 2, 4, 8$, respectively. Coarse pooling reverses or nearly cancels the signal; sibling pairs first preserve a positive intermediate-group signal, and full resolution removes grouping differences. The implemented unit-row normalization divides these sums by the square root of group size, preserving their signs. This budget follows from the design, not an independently discovered optimum (Fig. 3B).

Controls preserved rank while changing recipient assignment, support geometry or sparsity: cyclic derangement, depth-interleaving, random-sparse and dense random fields. The per-seed best-control summary takes their maximum after training; it is not a learned selector. A training-informed principal-component reference is retained separately (Methods).

Increasing bandwidth improved correct routing, while derangement caused large losses (Fig. 3C). Degree- and depth-preserving rewiring removed the task-aligned advantage at intermediate budgets but did not affect fully shared or fully resolved ancestry routes (Supplementary Section S3 and Source Data).

The ancestry-specific effect was much smaller. Ancestry performed worse than the best matched control at low budgets, exceeded it by 1.27 percentage points at $K = 4$, and tied at full resolution. The paired $K = 4$ interval was 0.59–1.95 points, with 15/20 positive differences; a paired Wilcoxon signed-rank test gave Holm-adjusted $P = 0.0101$ across four budgets. Individual control contrasts and paired seed differences distinguish this effect from the much larger correct-versus-deranged contrast (Fig. 3E,F).

Flat, grouped-point and gated-point representations receiving the same field behaved identically. Their equivalence locates the conditional advantage in feedback organization and supplies no independent confirmation or evidence for a particular physical carrier.

A separate twenty-seed cohort tested how supplied context cues could set the coefficients (Fig. 3D). An encoder trained on 256 calibration cues with noise SD 0.5 and their four-route activation targets improved task accuracy over a frozen profile by 7.16 points, but remained 54.47 points below oracle routing. Exploratory delivery of only its most probable route reduced this noisy-cue gap to 16.26 points. With sixteen noise-free cues, the correct route was already identified: soft coefficients still lost 60.90 points, whereas hard selection matched the oracle in all twenty seeds. Leakage into incompatible routes therefore mattered even with correct identification. Hard selection retained a deficit under noisy cues, and delaying either readout across independent contexts removed its benefit (Supplementary

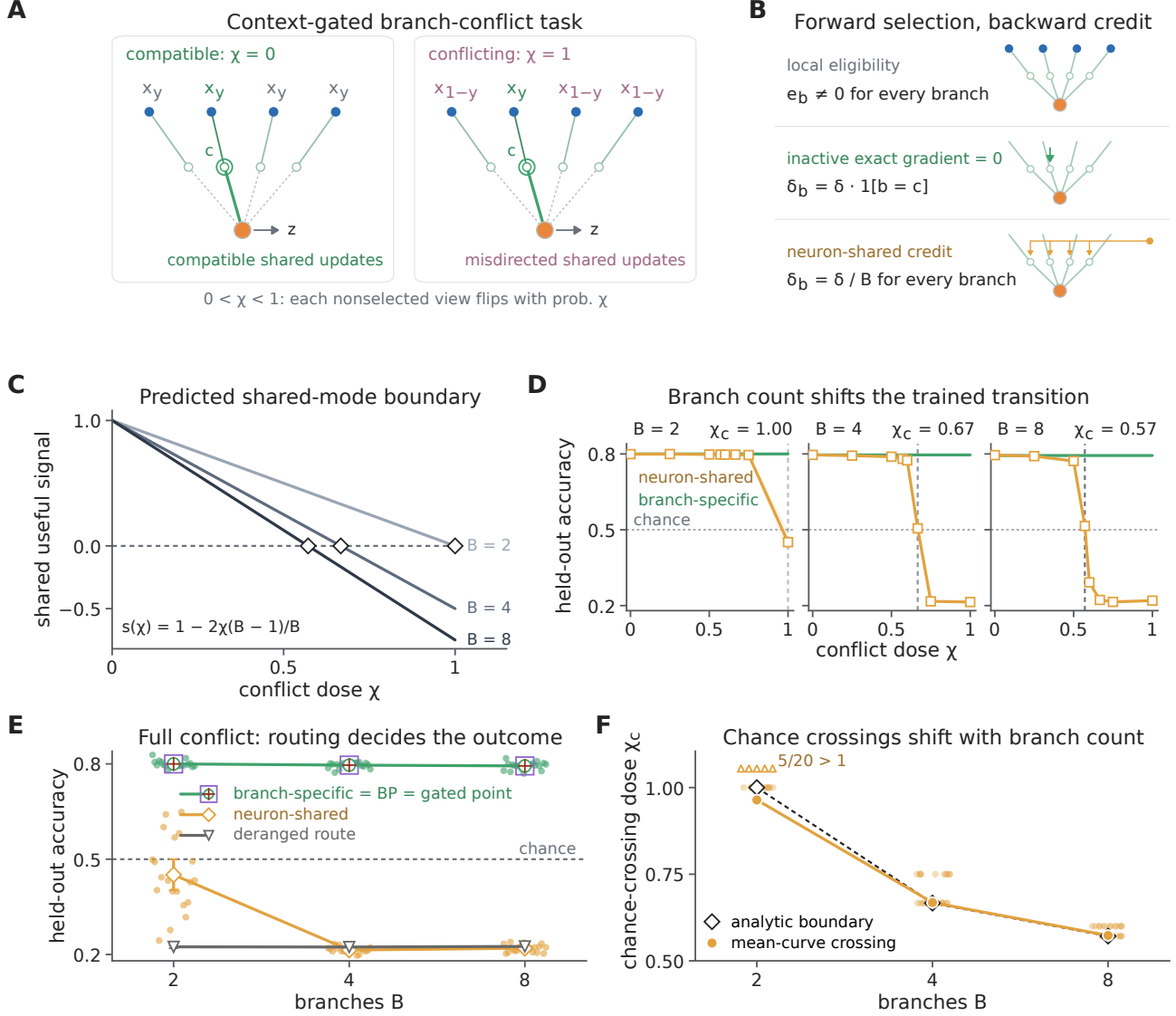


Figure 2: Branch selection prevents learning from inactive distractors. One logistic output combines B parallel branch weight vectors; there are no serial or conductance nonlinearities. **A**, Context-gated Fashion-MNIST task. Context c selects the branch determining logit z ; every branch receives an image and forms an ungated eligibility. Conflict dose χ independently replaces each nonselected image by an opposite-class image. Solid and dashed proximal paths are selected and gated-off forward routes; the double ring marks context. **B**, Selected feedback: $\delta_i \mathbf{1}[b = b_i^*]$, shared feedback: δ_i / B . Here δ_i is the output error, b_i^* the selected branch and $\mathbf{1}$ the indicator; unselected exact derivatives are zero. A local eligibility gate gives the same update. **C**, Centered symmetric-evidence prediction $s(\chi) = 1 - 2\chi(B - 1)/B$: shared-mode compatibility vanishes at $\chi_c = B/[2(B - 1)]$. This mean-field boundary is distinct from a trained chance crossing. **D**, Held-out accuracy over conflict dose. **E**, At full conflict, correct selection, backpropagation (BP) and the algebraically equivalent gated-point calculation coincide; cyclic derangement sends signals to the wrong branch at matched rank and sparsity. Selection-by-conflict interactions are positive in all 20 seeds for each B . **F**, Analytic threshold (diamonds), first interpolated crossing of mean accuracy (orange), and each seed's first sampled at-or-below-chance dose (faint dots). Open overflow triangles retain five $B = 2$ seeds that do not cross within the sweep. Curves and bars summarize 20 paired seeds with 95% seed-bootstrap intervals. Branch-count comparisons do not hold capacity fixed.

210 **Cross-branch interactions change the spatial credit that learning uses**

211 Branch interactions can change credit’s sign and magnitude between examples. We isolated this effect using targets
 212 with identical input-sensitivity spectra on the same compatible tree.

213 Eight independent inputs x_i were equally likely to be -1 or $+1$. Each branch computed $u = a + bu_L + cu_R + du_Lu_R$
 214 with four trainable coefficients. This multi-affine unit is affine in either child while the other is fixed. Pairwise
 215 targets summed four disjoint products, $f_M = \frac{1}{2} \sum_{(i,j) \in M} s_{ij} x_i x_j$; quartic targets combined complementary quartets,
 216 $f_A = \frac{1}{2} (s_A \prod_{i \in A} x_i + s_{\bar{A}} \prod_{i \notin A} x_i)$. Here M partitions inputs into pairs, A is one quartet, $\bar{A}$ its complement, and signs s
 217 were independently sampled. Within each seed, targets shared a compatible balanced tree, input assignment, initial
 218 weights and examples (Fig. 4A). Both have input-sensitivity second moment $\mathbb{E}[\nabla_x f \nabla_x f^T] = I_8/4$, giving identical
 219 strengths of independent sensitivity directions.

220 A scalar branch output cannot transmit several independently required input combinations to its parent. This
 221 communication constraint gives an exact compatibility criterion: at each subtree boundary, the matrix of interaction
 222 coefficients between inside and outside input products must have rank at most one after removing the constant inside
 223 row. The condition is necessary and sufficient across all boundaries; discarded singular values bound representation
 224 error (Methods; Supplementary Section S4).

225 Logical rules make this forward constraint concrete. All fifteen binary trees over four inputs can represent AND,
 226 OR and parity exactly, but mixed rules can require particular input groupings. The rule $(a \text{ AND } b) \text{ OR } (c \text{ AND } d)$
 227 needs only two serial stages, whereas $a \text{ AND } [b \text{ OR } (c \text{ AND } d)]$ needs three in this scalar model, despite containing the
 228 same two AND gates and one OR gate. These representability statements are independent of training (Supplementary
 229 Figs. S13–S14). They identify possible morphologies before testing credit-dependent learning.

230 All credit rules used the same exact local eligibilities and scalar output error. We compared exact paths with
 231 unit broadcast and a fixed six-site profile calibrated from each student’s mean path derivatives over 256 unlabeled
 232 initialization examples. Three development seeds selected learning rates with equal budgets; twenty fresh paired
 233 seeds tested the frozen choices. A separate nested target summed products over prefixes of two, four, six and eight
 234 inputs on its own compatible tree.

235 The calibrated profile learned pairwise targets but remained inaccurate on quartic and nested targets (Fig. 4B–D).
 236 At the primary 1,024-step checkpoint, mean normalized mean squared errors (NMSE) for exact and calibrated credit
 237 were 0.0317 and 0.0236 on pairwise targets, 0.1516 and 0.9813 on quartic targets, and 0.0230 and 0.7690 on nested
 238 targets. NMSE divides squared prediction error by clean target variance. The predefined quartic-minus-pairwise
 239 difference in calibrated-minus-exact gaps was 0.838 (95% paired interval, 0.719–0.931), positive in all twenty seeds;
 240 a common learning rate gave 0.843 (0.727–0.935). Exact quartic learning was heterogeneous at this checkpoint:
 241 seventeen fits were near the noise floor and three remained stalled. The lower pairwise mean for calibrated credit
 242 likewise reflected one slow exact fit; exact credit performed better in fifteen of twenty seeds (Supplementary
 243 Figs. S15–S16).

244 A balanced continuation of every algebraic trajectory to 16,384 updates tested whether these gaps persisted
 245 without retuning. All twenty selected-rate Adam exact quartic fits reached the noise floor by 4,096 updates, and the
 246 pairwise mean lag disappeared. At 16,384 updates, the quartic-minus-pairwise calibrated-credit deficit was 0.952
 247 (95% paired interval, 0.883–1.012) at terminal states and 0.790 (0.691–0.880) under validation-based selection,
 248 positive in all twenty seeds (Fig. 4E). Fixed-profile arms contacted coefficient bounds; exact arms did not. These are
 249 the same previously observed seeds and bounded dynamics, with common-rate, per-seed and optimizer controls
 250 (Supplementary Figs. S15–S16 and Table S8).

251 The trained fields reveal a distinction missed by the input spectrum. Let $\mathbf{q}(x)$ contain the six nonsomatic
 252 derivatives of root output with respect to branch output; exact branch credit is scalar output error times $\mathbf{q}$. At the
 253 1,024-step checkpoint, allowing an oracle amplitude per example, the best fixed profile captured 99.7% of exact-
 254 trained path-field energy for pairwise targets, but 46.1% for quartic and 44.8% for nested targets. The separation
 255 persisted at 16,384 updates: capture was approximately 100%, 39.6% and 40.5%, respectively (Fig. 4F). Loss-
 256 error weighting, eligibility weighting, site-normalized spectra and replay of 600 earlier fits preserved the original

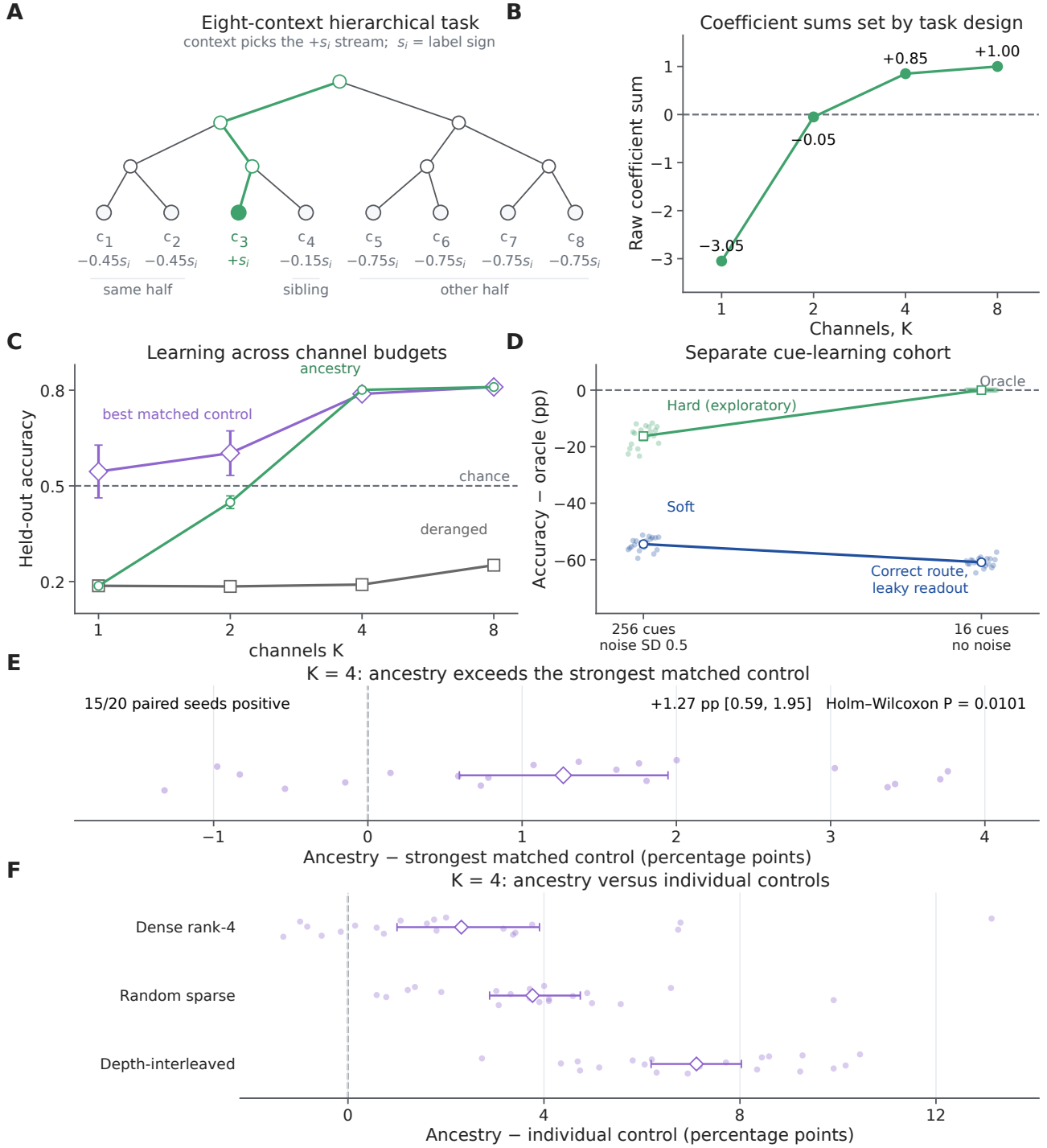


Figure 3: **Hierarchical distractors favor ancestry at intermediate bandwidth.** **A**, The task-selected stream carries label sign s_i ; distractor coefficients are shown on the tree. **B**, Raw class-signal sums within cued ancestry groups. Unit-row normalization divides by square root of group size, preserving the designed sign change. **C**, Accuracy for ancestry, derangement and the per-seed maximum over four matched controls, computed after training. Dashed line: chance. **D**, Separate coefficient-learning cohort: soft and maximum-probability hard readouts relative to oracle routes, with supplied calibration cues and route-activation targets. Twenty seeds are paired within each condition. The noisy soft comparison is primary; hard readout and the noiseless condition are exploratory. Calibration size and cue noise differ between conditions. **E**, Four-route ancestry-minus-best-control differences; Holm-adjusted two-sided Wilcoxon $P = 0.0101$ across four budgets. **F**, Dense, sparse and depth-interleaved control contrasts on a separate scale. Symbols and bars are means and 95% seed-bootstrap intervals; small points in **D–F** show twenty paired seeds in each cohort.

separation (Supplementary Section S4). Uniform-profile capture did not: a nearly one-dimensional field can have a signed, nonuniform direction. Positive fixed-profile scaling largely cancels in Adam, explaining calibrated/sign-only agreement; it does not explain the slower pairwise learning with changing exact paths.

The learned credit spectra help explain these input-matched targets' different tolerance of fixed profiles. At fixed weights and inputs, path derivatives are target-independent; their spectrum describes learned states and coordinates, not a task-only initialization diagnostic. The results favor example-dependent spatial credit without establishing that all six exact channels are necessary.

Additional controls delimit the result. Fixed-profile Adam learned a compatible positive-conductance teacher accurately; broadcast learned aligned XOR-of-AND, whereas parity favored exact credit under the tested optimizer (Supplementary Figs. S13–S14 and S17). Noisy-query interaction estimation also yielded useful trees, retained as supporting structure-design evidence in Supplementary Fig. S35. Interaction structure therefore motivates specific credit comparisons rather than a universal monotonic resolution law.

Context-dependent branch tuning makes spatial credit useful in conductance trees

Conductance trees provide a biophysical model for testing this distinction. All rules learned a seven-compartment teacher with monotonic branch responses: extended Adam training kept broadcast NMSE below 0.0006 despite a varying path field (Supplementary Fig. S18). A separate construction tested context-selected branches with aligned or opposing feature tuning (Fig. 5A,B).

Four terminal compartments fed two proximal compartments and a soma, using the directed E/I conductance balance (Methods). Both subtrees received the same positive feature pairs, $\exp(z_j)$ and $\exp(-z_j)$, for two independent latent inputs z_j . A binary context increased inhibitory conductance in one proximal compartment. Teacher excitation and inhibition produced increasing feature responses in both subtrees for the aligned task, but increasing responses in one and decreasing responses in the other for the opposed task. Teacher and student shared the circuit, ensuring representability. Students shared the teacher's nominal proximal and coupling gains, with independent Gaussian perturbations of SD 0.4 in log conductance; terminal gains started near one. All 24 conductances were plastic, and paired conditions shared initial weights and example streams.

At fixed weights, binary context generates at most two exact path profiles. Their gains are positive: context changes the relative weighting of the two distal subtrees, while proximal credit is context invariant. The nominal parent's conductance denominator changes by approximately eightfold. Thus branch selection is the relevant distinction, connecting this task to Fig. 2.

To supply selection without oracle coefficients, a local rule delivered the somatic error to terminal parameters only when their parent was uninhibited, while retaining ungated error at proximal compartments and the soma. It used the supplied inhibitory context and exact local eligibilities, without evaluating exact paths or projecting an exact field. Two fixed distal profiles with oracle amplitudes and unit proximal credit supplied a separate capacity control. An additional proximal profile in the earlier three-profile control therefore addresses proximal updates, rather than another context distinction.

Twenty new paired seeds tested the rule under a protocol frozen after exploratory results. All nine rules ran at three common rates through 16,384 updates; the primary rate and 4,096-update window were fixed before these outcomes. On opposed targets, mean held-out NMSE was 2.18×10^{-5} for exact credit, 2.33×10^{-5} for the local gate, 2.28×10^{-5} for the two-profile oracle with unit proximal credit, and 0.487 for broadcast (Fig. 5C,D). On aligned targets, the corresponding exact, local-gate and broadcast errors were 7.53×10^{-6} , 1.06×10^{-5} and 5.53×10^{-4} . The opposed-minus-aligned difference in broadcast-minus-gate error was 0.487 (95% paired interval, 0.360–0.615), positive in all twenty seeds. It remained 0.468 (0.332–0.604) within the longer window (Fig. 5E).

The controls identify what the gate must preserve. Reversing its distal selection gave opposed-task NMSE 0.949. Applying it at proximal compartments as well gave 0.123 and prevented both proximal inhibitory gains from learning: their eligibility is nonzero only during inhibition, precisely when that gate closes (Fig. 5F). The proper local rule changed all 24 parameters and succeeded across the three tested rates. A continuous shunt-proportional distal gate reached NMSE 2.32×10^{-5} under the same primary rate and budget (Fig. 5F). Both gates were tested with binary inhibitory contexts; continuously varying or noisy context remains untested.

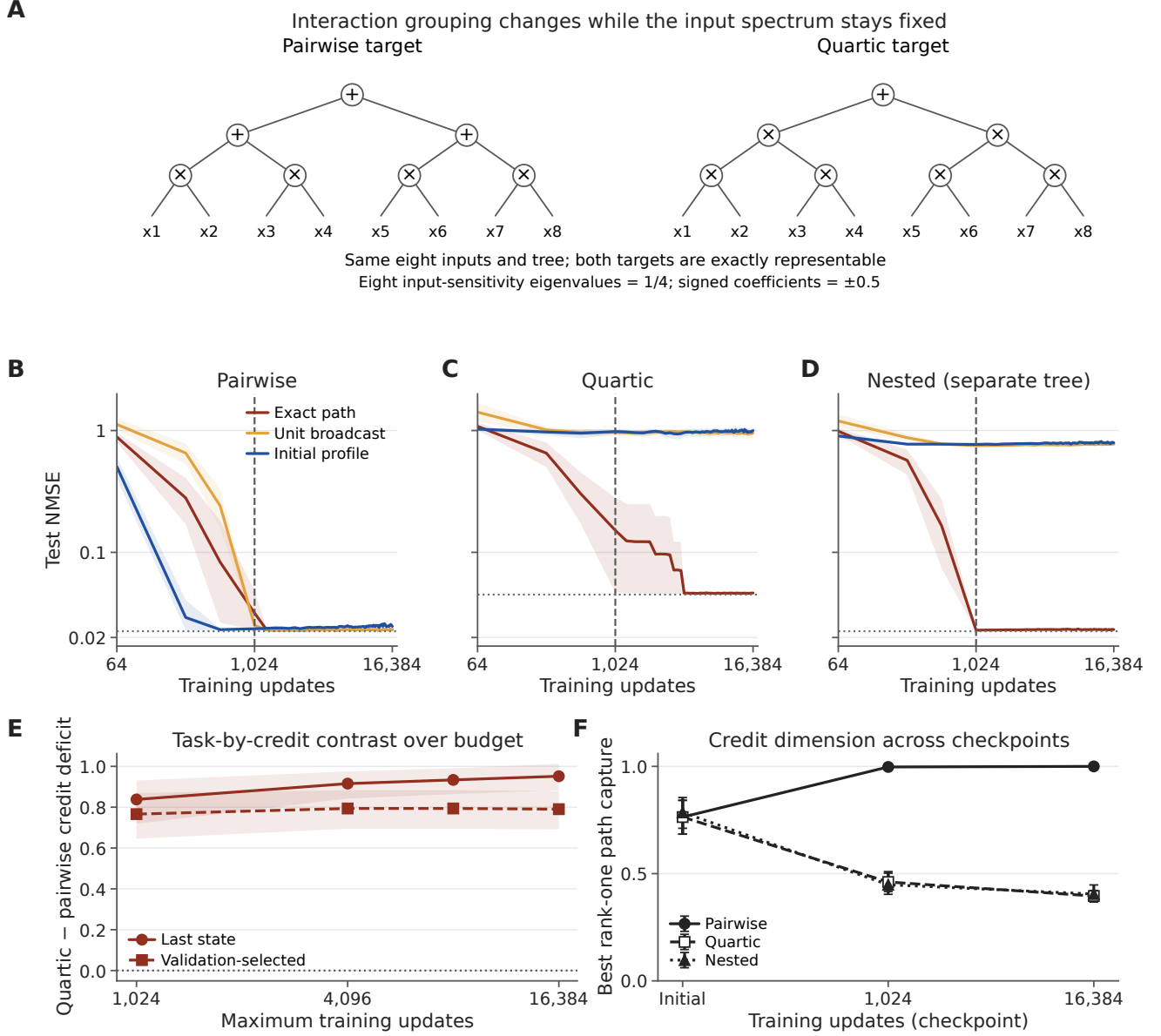


Figure 4: Matched input spectra yield different learned credit geometry and tolerance of fixed profiles. **A**, Pairwise and quartic targets share eight inputs, a compatible tree, initialization and examples. Signed coefficients have magnitude 0.5; both targets have input-sensitivity second moment $I_8/4$. Nodes illustrate interaction supports. **B–D**, Held-out normalized mean squared error (NMSE) for exact credit, unit broadcast and an initial profile calibrated on 256 unlabeled examples. Twenty seed blocks continue to 16,384 updates at the original development-selected Adam rates and coefficient bounds. Nested targets use a separate compatible tree. Curves are current-state means; shading shows pointwise descriptive 95% whole-seed intervals. Vertical lines mark the primary 1,024-step checkpoint; horizontal lines mark noise-only NMSE (0.0225 for pairwise/nested, 0.045 for quartic). **E**, Quartic-minus-pairwise difference in calibrated-minus-exact NMSE at current states and minimum-validation-error states within each budget, with paired 95% intervals. These continuations use previously observed seeds without retuning. **F**, Best rank-one capture of six-site exact path-field energy at initialization, 1,024 and 16,384 updates, with twenty-seed 95% intervals. The primary and continued checkpoints remain distinct; spatial directions and example-specific amplitudes are oracle fits.

305 The gate coarsens the exact rule’s strong context-dependent gain ratio. Same-state diagnostics showed that
 306 broadcast mixes opposing context-conditioned updates, despite nonnegative per-example alignment with exact
 307 gradients (Supplementary Section S5 and Source Data). This supplies a local implementation given inhibitory
 308 context and somatic error, while leaving their biological origin open. The comparisons remain budget- and parameter-
 309 bounded. Exact and local-gate fits did not reach conductance bounds, whereas most opposed broadcast fits did; the
 310 earlier wide-bound and rate controls remain in Supplementary Fig. S19.

311 Task organization determines the benefit of serial dendritic computation

312 The preceding comparisons hold a compatible tree fixed. Varying serial structure instead tests whether successive
 313 compartments help remove unwanted multiplicative changes in input strength. Physical depth D_p counts forward
 314 stages, whereas feedback resolution counts distinctions in the backward field.

315 Each network contained 64 somata and eight nonsomatic compartments per soma. D1 [8], D2 [2, 3] and D3
 316 [2, 1, 2] distributed those compartments across one to three stages in the three- and four-tier comparisons while
 317 preserving contacts and trainable resources. The two-tier D2 tree used [4, 1], also with eight compartments. Branching
 318 vectors are ordered from soma to tips. A resource-identical grouped-point control connected all eight modules
 319 directly to the soma; raw-additive trees removed divisive conductance normalization. Flexible point networks
 320 provided additional, approximately parameter-matched comparisons.

321 Distal excitation carried class information multiplied by nuisance gains. Inhibitory sensors reported those gains
 322 with alignment α : zero retained the sensor distribution but removed trial-specific information, and one reported the
 323 factor exactly. With inhibitory reversal zero, the sensors entered the conductance denominator and could support
 324 successive divisions [24, 41]. Nested factors acted over fine, coarse and global input groups; flat factors had
 325 equal-resolution supports; locally cancellable factors placed the needed ratio within each module (Fig. 6A). Both
 326 nested and flat targets used commutative products; they differed in where factor information was available.

327 In the 180-epoch hierarchy comparison, the two-tier task favored D2, whereas three- and four-tier tasks favored
 328 D3 among the tested architectures (Fig. 6B). The grid was incomplete, and its largest tested means identify budget-
 329 specific outcomes rather than a rule equating task tiers with optimal depth. Task-family controls showed what serial
 330 organization contributed: at full sensor alignment, exact-BP serial trees exceeded the resource-identical grouped-
 331 point control by 30.32 percentage points for nested gains and 22.20 points for flat gains, but had no advantage for
 332 locally available ratios. Removing trial-specific sensor information eliminated these large benefits (Fig. 6C). Serial
 333 structure was useful when the task required integrating factor information distributed across modules.

334 Credit comparisons within this useful forward architecture depended on training duration. Ten paired seeds
 335 compared D1 exact BP with five D3 credit/optimization conditions for up to 600 epochs, retaining the original rates,
 336 parameter groups and thirty-epoch validation patience. SI reports restart agreement and complete validation histories.
 337 At validation-selected states from the same trajectories, exact-BP D3 exceeded D1 by 30.82 points at 180 epochs and
 338 38.17 at 600 (Fig. 6D). Eight of ten D1 references stopped early; all fifty D3 fits reached the cap with validation loss
 339 still falling. The forward benefit is therefore budget-specific, with unequal optimization maturity across depths.

340 Within D3, the LocalCA exact-path-minus-shared-soma accuracy gap changed from +10.86 points at 180 epochs
 341 to -1.52 at 600 (95% paired interval, -1.83 to -1.23), with this budget-indexed sign change in all ten seeds. The
 342 mean gap remained negative from epoch 315, reached -2.94 points at epoch 486, and moved back towards zero by
 343 600 (Fig. 6E). Exact-path LocalCA nevertheless retained lower mean cross-entropy over epochs 180–600, ending at
 344 0.254 versus 0.276 (Fig. 6F). Accuracy counts threshold crossings, whereas cross-entropy also reflects predicted
 345 probabilities. Under the BP recipe, the 600-epoch exact-minus-broadcast accuracy gap was 0.48 points (0.40–0.55).
 346 Thus serial organization and spatial credit answer distinct questions: distributed factor information supports the
 347 forward architecture, while the eventual ranking of the tested credit rules remains unresolved (Supplementary
 348 Figs. S21–S23 and Table S9).

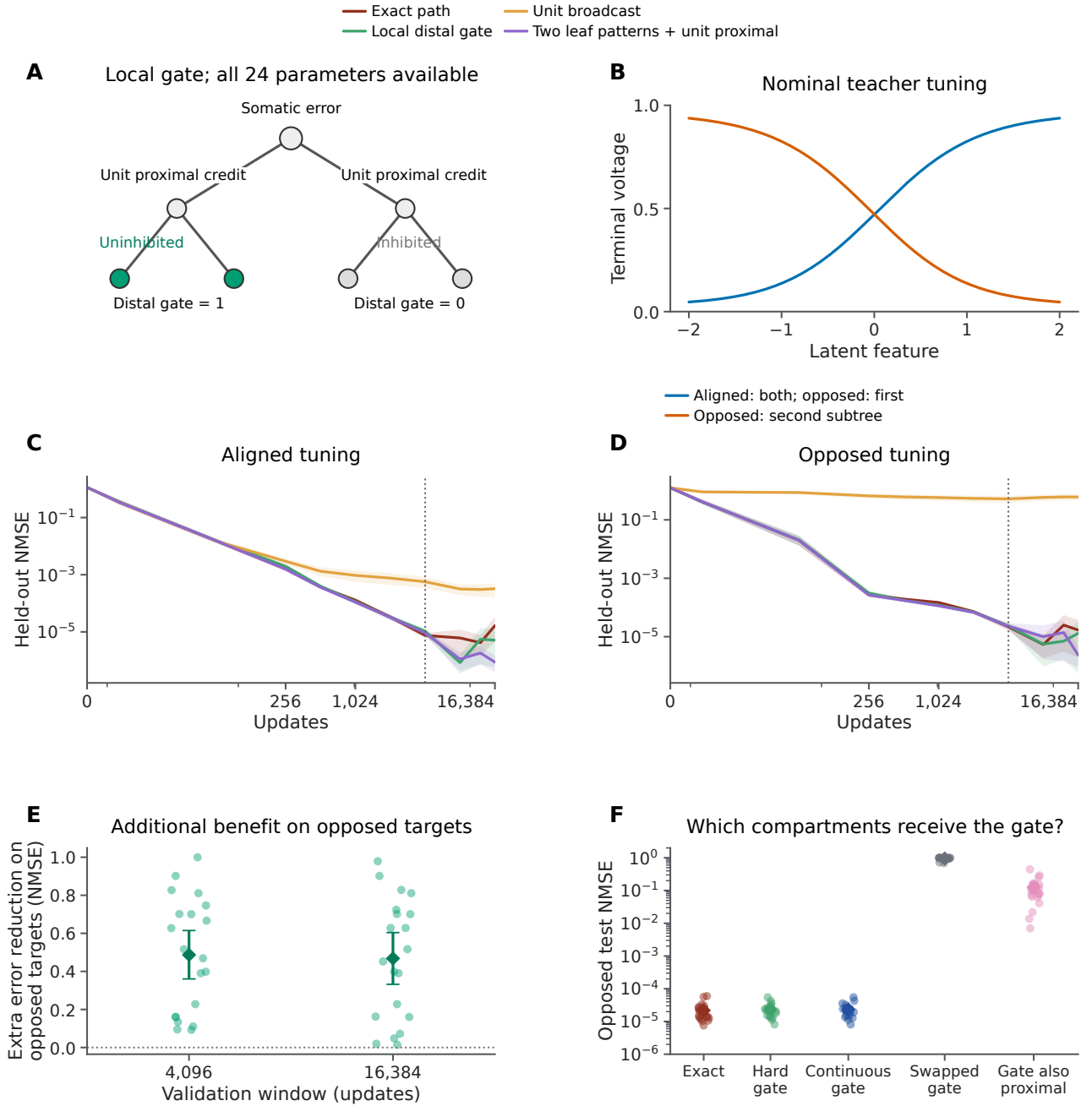


Figure 5: A local inhibitory gate supplies useful branch selection in a conductance tree. **A**, Four terminal compartments feed two proximal compartments and a soma. All 24 conductances are plastic. The local rule gates terminal credit by the parent’s inhibitory activity while retaining unit multipliers for proximal and somatic credit. Local eligibilities are unchanged. **B**, Nominal teacher tuning for aligned and opposed branches; both tasks receive identical positive feature pairs. **C,D**, Fixed-checkpoint test NMSE for aligned and opposed targets, respectively. Twenty new paired seeds share initial students and example streams across rules. Adam rate 0.03 and both budgets were fixed before execution. The two-profile control uses oracle distal amplitudes and unit proximal credit. Dotted lines mark 4,096 updates; every trajectory continues to 16,384. Update axes are linear through 64 and logarithmic thereafter. **E**, Opposed-minus-aligned difference in broadcast-minus-local-gate NMSE, evaluated at minimum-validation-error states within each budget. **F**, Opposed-task outcomes within the primary window for exact credit, hard and continuous distal gates, swapped selection and proximal gating. The continuous gate uses $1/(1 + g_p^I a_p)$ with the same binary inhibitory activity a_p as the hard rule. Gating proximal credit freezes two inhibitory gains. Points show every seed; larger symbols and bands in **C–F** show means and descriptive 95% whole-seed bootstrap intervals. All rates and nine rules are retained in Supplementary Fig. S20. Earlier exploratory gate runs are excluded.

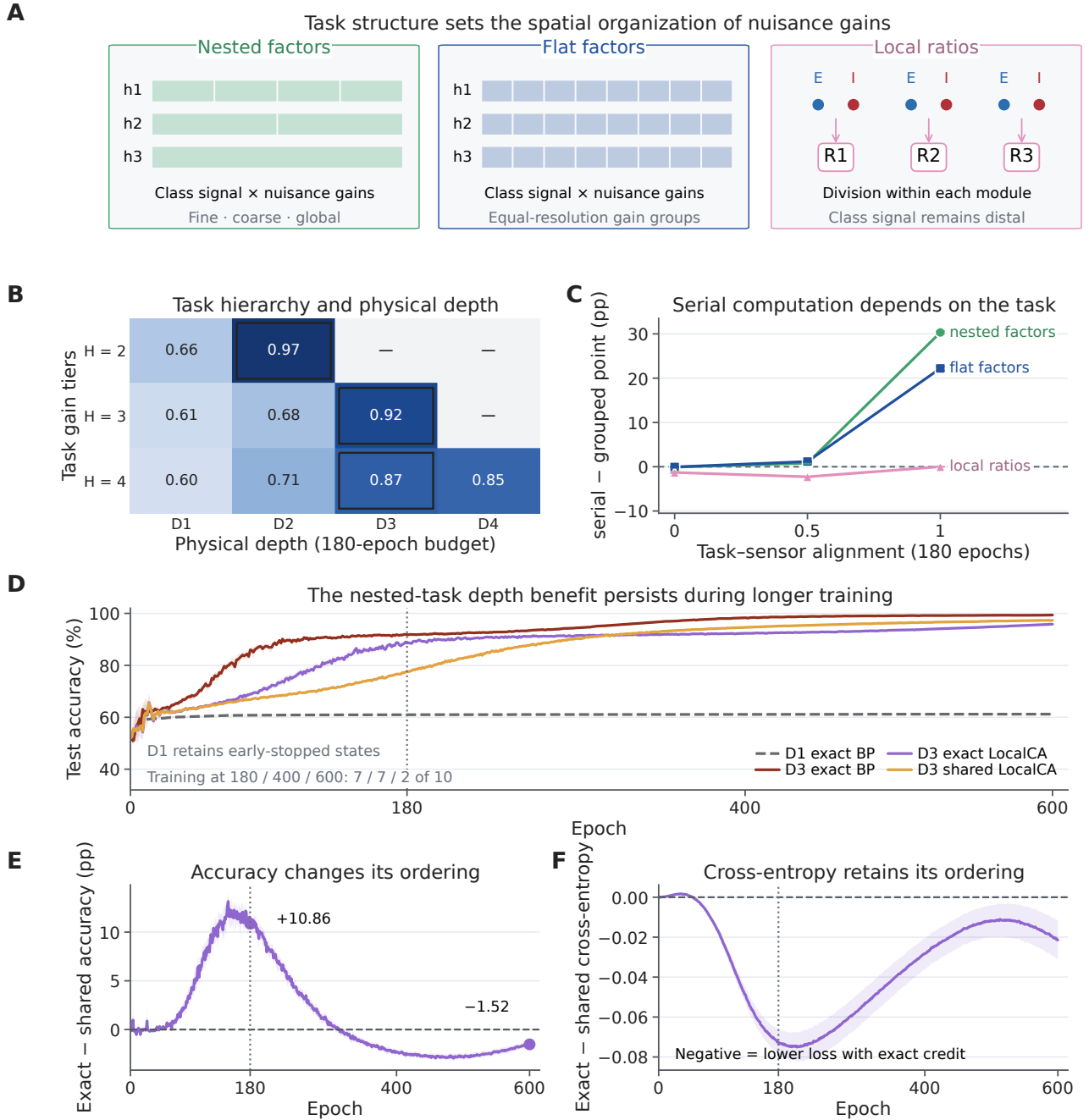


Figure 6: Task organization determines when serial dendritic computation helps. **A**, Nested gains have fine, coarse and global supports; flat gains have equal-resolution supports; local-ratio tasks expose division within each module. Drawings indicate input access, not exact cancellation. E/I, excitatory/inhibitory inputs. **B**, Mean exact-BP accuracy from ten seeds per tested task gain-tier H and physical-depth D combination at 180 epochs. Outlines mark the largest tested means; dashes denote untested combinations. The incomplete grid does not establish converged depth optima. **C**, Exact-BP serial-minus-resource-identical-grouped-point accuracy across task families and sensor alignment, with ten paired seeds and 95% intervals at 180 epochs. **D**, Nested-task D1 [8] and D3 [2, 1, 2] trajectories at matched resources. Validation-selected states are carried forward after stopping. The annotation counts D1 fits still training: seven of ten at epochs 180 and 400, and two at 600. **E,F**, Paired D3 exact-minus-shared LocalCA differences in accuracy and cross-entropy. Accuracy changes sign between 180 and 600 epochs; exact credit retains lower mean cross-entropy. All fifty D3 fits reach the 600-epoch cap with falling validation loss; eight of ten D1 references stop earlier. Shading in **D–F** shows pointwise descriptive 95% intervals over ten paired seeds. Both recipes use Adam with their original rates and parameter groups; validation selection does not establish convergence.

Reconstructed arbors provide spatial routes beyond a shared broadcast

The learning tasks motivate asking what delivery patterns real arbors supply. MICrONS links reconstructed anatomy and connectivity to visual responses [42]. Observed skeletons, soma positions and mapped contact locations and areas define the geometry; conductances, voltages and gradient fields are modeled.

Each inhibitory-bearing segment supplied a candidate route over its input-bearing descendants (Fig. 7A). These supports form dictionary columns, supplemented by a constant column for common broadcast (Fig. 7B). For a site-by-route matrix A_K , diagonal excitatory-area weights W and weighted least-squares projector $P_{A_K, W}$, capture of a field $\mathbf{t}$ was

$$\mathcal{C}(A_K; \mathbf{t}) = \frac{\|P_{A_K, W} \mathbf{t}\|_W^2}{\|\mathbf{t}\|_W^2}, \quad \|\mathbf{v}\|_W^2 = \mathbf{v}^T W \mathbf{v}. \quad (7)$$

For an ensemble of perturbations, numerator and denominator sum over field columns. A common-constrained SVD dictionary provided a ceiling at the same column budget. Controls used random anatomical routes, cable-distance bins, site-shuffled routes preserving nonzero counts, and surrogate trees preserving degree and depth while changing ancestry.

Small focal shunts in reciprocal cables generated gradient-change fields without inspecting the selected dictionary. These fields nevertheless favor ancestry by construction under a somatic loss, as derived below. The test asks how a small dictionary compresses the tree’s own passive responses. In the initial eight cells, broadcast captured 31.4% of field energy, near the 32.6% rank-one ceiling. Every dictionary therefore contained one common broadcast and $K - 1$ spatial profiles, with the protocol fixed before evaluation in 47 disjoint cells from the same mouse and a reconstructed second mouse.

Varying K separates the common signal from the additional spatial information each dictionary can capture (Fig. 7C). At eight columns in the 47-cell cohort, broadcast-plus-ancestry captured 0.619 of total energy versus 0.578 for degree–depth surrogates. The paired advantage was 0.041 (95% cell-bootstrap interval, 0.020–0.062), positive in 38 of 47 cells. This mean conceals heterogeneity: in 11 cells, more than half of the 200 surrogates matched or exceeded the actual tree. Depth bins captured 0.511 and site-shuffled profiles 0.436; the ancestry-minus-depth difference was 0.108 (0.077–0.135), positive in 45 cells. After removing broadcast, ancestry and depth-bin residual capture was 0.527 and 0.392. Residual contrasts isolate the spatial advantage (Fig. 7D). Actual ancestry improves average compression within this passive response family, while coarse geometry explains much of the capacity.

Capture and delivery cost are separate constraints (Fig. 7E). The eight-column ancestry dictionary used 18.8% of the entries of a fully dense dictionary on average; its site-shuffled control preserved exactly the same nonzero count. Equal column count does not ensure equal rank, and nonzero entries are a delivery-cost proxy rather than feedback synapses or metabolic expenditure.

The ordering held in the initial eight cells (Fig. 7F). In the Pinky second mouse, eight of ten qualifying cells supported the eight-column comparison. Their 9–13 input-bearing sites put ancestry near the residual-capture ceiling, so lower-budget comparisons and exclusions remain separate from the larger arbors. Direct E/I labels covered 4.71% of disjoint-cohort inputs; the initial cohort’s 73.0% direct-versus-proxy agreement concerned jointly labeled contacts. Geometry, labeling and compression sensitivities accompany the cohort controls in SI (Supplementary Figs. S24–S27).

Focal shunting sets gains on ancestry-defined credit routes

A focal shunt provides an exact link between anatomical routes and conductance-dependent gains. We compared a focal inhibitory shunt with a current injection matching its baseline focal current in reconstructed reciprocal cables. A separately solved somatic current restored baseline somatic voltage after either intervention. The endpoint was the change in the modeled excitatory-conductance gradient.

Let G be the passive conductance matrix and $\mathbf{b}$ the current-source vector, so that baseline voltages satisfy $G\mathbf{V} = \mathbf{b}$. With $\mathbf{e}_k$ the unit vector at compartment k , adding inhibitory conductance κ_k at compartment k changes both G and its source: $G' = G + \kappa_k \mathbf{e}_k \mathbf{e}_k^T$, $\mathbf{b}' = \mathbf{b} + \kappa_k E_1 \mathbf{e}_k$. The matched injection supplies baseline current $\kappa_k (E_1 - V_k)$ without changing G . The adjoint $\mathbf{q}$ gives loss sensitivity to current injected at each compartment. For a fixed somatic loss

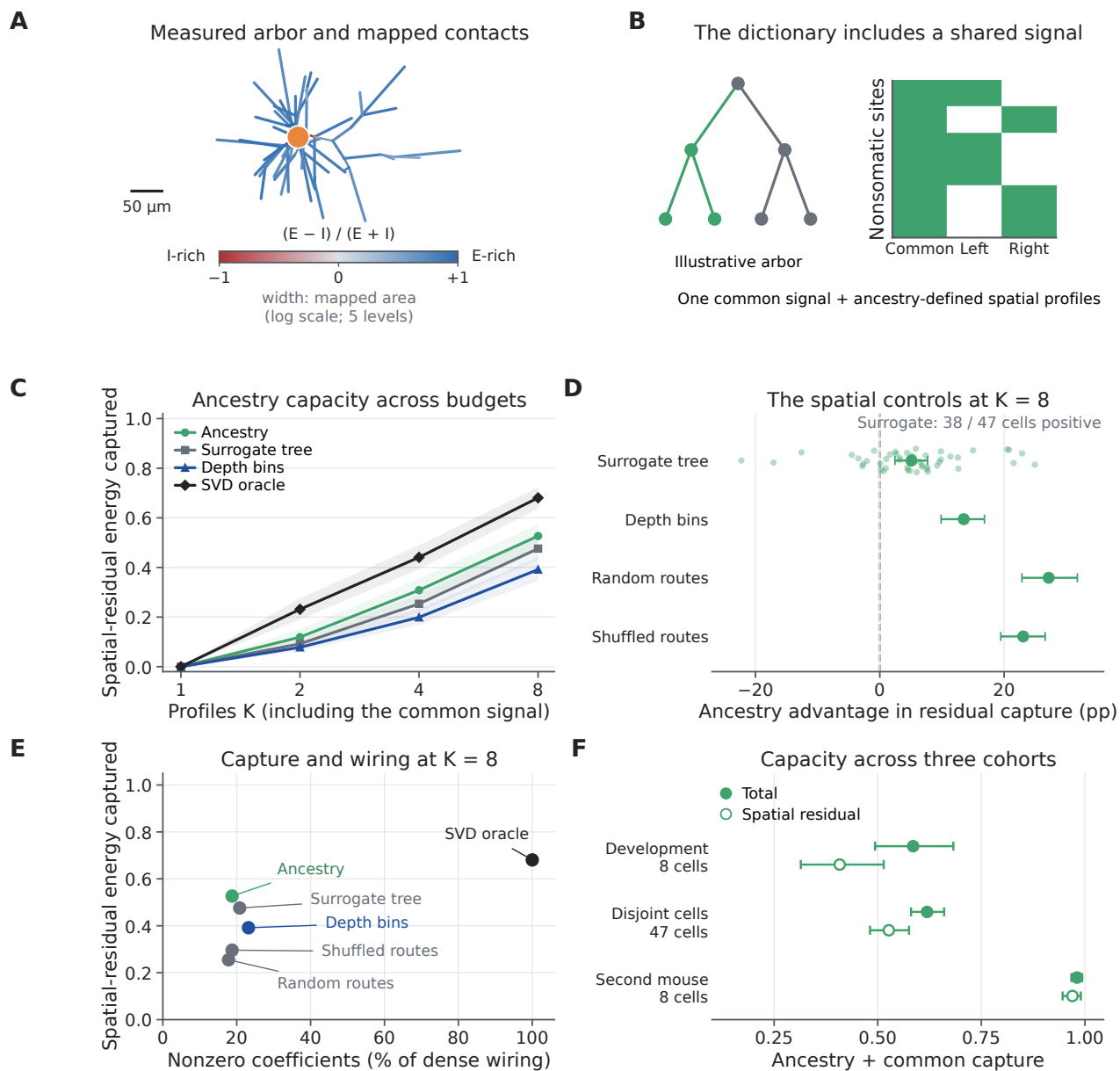


Figure 7: Anatomy supplies sparse spatial dictionaries beyond a shared broadcast. **A**, Median-sized arbor from the initial eight-cell cohort. Color represents mapped E/I contact-area balance; width represents total contact area. Scale bar, 50 μm . **B**, Illustrative subtree-to-column mapping with a shared constant profile. **C**, Capture of field energy remaining after a weighted common projection at $K = 1, 2, 4, 8$, each evaluated in all 47 disjoint v661 cells. Every dictionary contains the same constant profile; the common-constrained SVD is an oracle ceiling. **D**, Paired ancestry advantages at $K = 8$. Surrogate trees preserve segment depths and parent out-degrees; shuffling preserves column nonzero counts. Randomized controls average 200 draws per cell. Small points on the surrogate comparison show all 47 within-cell differences: 38 positive and nine negative. **E**, Residual capture versus mean nonzero coefficient density, relative to dense eight-column wiring. This is a delivery-cost proxy; equal K does not equate rank or wiring. **F**, Total and residual ancestry capture in the initial, disjoint and second-mouse cohorts. Pinky's eight eligible cells have only 9–13 input-bearing sites. Symbols and intervals in **C–F** summarize cell means and descriptive 95% cell-bootstrap intervals where shown. Capture uses squared excitatory-area-weighted energy. Passive shunt fields inherit ancestry structure through cable physics; this is anatomical compression capacity, not observed teaching.

394 derivative, it changes by

$$\mathbf{q}' = \mathbf{q} - \frac{\kappa_k \mathbf{q}_k}{1 + \kappa_k (G^{-1})_{kk}} G^{-1} \mathbf{e}_k. \quad (8)$$

395 The column $G^{-1} \mathbf{e}_k$ gives the voltage response to unit current at k . On a passive tree with a somatic loss, the
 396 resulting adjoint change has an exact ancestry structure: all descendants of the shunt share the attenuation factor
 397 $[1 + \kappa_k (G^{-1})_{kk}]^{-1}$, while each sister block along the path to the soma shares its own factor (Fig. 8A; Supplementary
 398 Section S8). These disjoint blocks form an ancestry partition. Weighting their indicators by the baseline soma-to-site
 399 transfer gives a dictionary on which the shunt acts through diagonal gains, providing a physical instance of Γ_u . The
 400 diagonal form applies in this partition basis; it need not hold for an arbitrary selected set of overlapping routes.

401 In the small-perturbation fields used for Fig. 7, the driving-force term was substantial: its weighted squared
 402 energy ranged from 0.99 to 3.50 times that of the full field across the eight initial cells. It opposed the adjoint term
 403 (weighted correlations -0.979 to -0.913). Despite this cancellation, 97.1–99.9% of full-field energy remained
 404 within each focal ancestry partition. The small outside-partition residual therefore does not imply a negligible
 405 driving-force contribution (Supplementary Section S8).

406 We compared descendants $D(k)$ with depth-matched off-route sites $C(k)$; Supplementary Section S8 gives
 407 a kernel criterion for stronger absolute adjoint changes below a shunt, with within-cell, reversal-potential and
 408 contact-label controls in Supplementary Fig. S28B,C. The analyzed gradient was sensitivity to open excitatory
 409 conductance, $\gamma_i = q_i (E_E - V_i)$. The local driving force $E_E - V_i$ can change differently within each block, so the
 410 complete synaptic gradient need not share a single block gain. With primes denoting post-intervention values and
 411 ε_γ a numerical floor defined in Methods, we measured absolute log change, $m_i = |\log[(|\gamma'_i| + \varepsilon_\gamma)/(|\gamma_i| + \varepsilon_\gamma)]|$, and
 412 localization $\Lambda_k = \text{median}_{D(k)} m_i - \text{median}_{C(k)} m_i$. Positive localization means a larger gradient change below the
 413 shunt.

414 In the permissive normalized model, shunts changed descendant gradients more strongly than depth-matched
 415 off-route gradients, with increasing localization at higher dose (Fig. 8B; Supplementary Fig. S28D). Current injection
 416 changed descendant gradients roughly threefold less. Its off-route changes were zero by construction under the
 417 soma-restoring control; this is not a general claim that current injection cannot change a gradient. Holding the
 418 post-shunt voltage but replacing its adjoint with the baseline adjoint removed most localization, supporting a transport
 419 contribution beyond the voltage change alone (Fig. 8C).

420 Physical calibration constrained the spatial selectivity of the gradient change. At axial resistivity $R_a = 150 \Omega \text{cm}$
 421 and specific membrane resistance $R_m = 15,000 \Omega \text{cm}^2$, mean shunt-minus-current localization was small and opposite-
 422 signed in the initial eight cells and a disjoint 45-cell cohort from the same mouse: -0.0019 and $+0.0024$, respectively.
 423 Signed changes showed attenuation of both descendant and off-route gradients at high resistance; lowering resistance
 424 primarily increased descendant attenuation (Fig. 8D). This increased the localization contrast by more than an order
 425 of magnitude (Fig. 8E). Distributed background conductance produced the same qualitative recovery at fixed high
 426 membrane resistance (Supplementary Section S8 and Source Data), consistent with a dependence on electrotonic
 427 state [25, 43].

428 Adding weak voltage-dependent channel currents produced a dose response close to the passive response at
 429 $R_m = 1,000 \Omega \text{cm}^2$ with one baseline-leak unit of distributed background conductance (Supplementary Section S8;
 430 Fig. S29D,E). This checks the local linearization around solved steady states. The baseline Jacobian included gate
 431 derivatives but remained fixed across doses, so the check does not establish robustness to strongly active dendrites or
 432 channel kinetics.

433 The partition identity and electrical calibration connect anatomical dictionaries to conductance-dependent route
 434 gains [26]. Their selectivity depends on cable state. These imposed perturbations establish modeled transport, not
 435 endogenous error production or plasticity.

436 Measured responses bound the anatomical alignment proposal

437 Measured responses test whether functional relationships follow the ancestry structure suggested by modeled capacity
 438 and transport. Seven MICrONS targets had complete arbors and imaged partners: 69 partners in the selected-scan

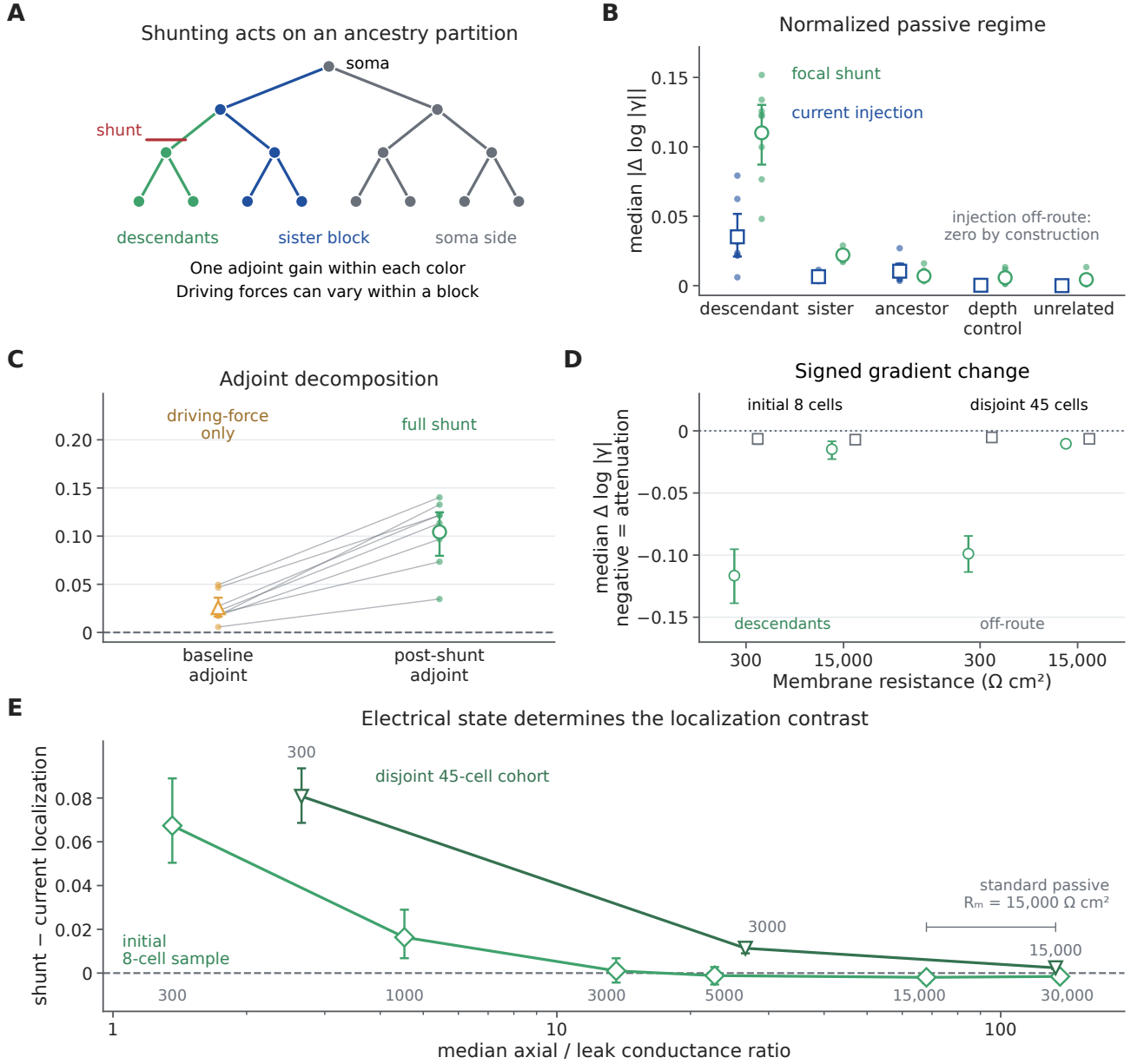


Figure 8: Focal shunts set ancestry-defined gains whose selectivity depends on electrical state. **A**, On a passive tree with a fixed somatic loss derivative, a shunt gives the same adjoint attenuation factor to every site in each colored ancestry block. The resulting gain is diagonal in the baseline-weighted partition dictionary; local synaptic driving forces can still vary within a block. **B**, Absolute log change in excitatory-conductance gradient magnitude by tree relation at unit dose, in the normalized passive regime for eight reconstructed cells. A current injection matches baseline focal current, and a separately solved somatic current restores baseline somatic voltage after either intervention. Off-route injection zeros arise from this construction. **C**, Keeping the post-shunt voltage while restoring the baseline adjoint removes most localization. Lines pair cells. **D**, Signed log change in gradient magnitude at the two physical-calibration endpoints, with $R_a = 150 \Omega \text{ cm}$ and $R_m = 300$ or $15,000 \Omega \text{ cm}^2$. Circles and squares show descendants and depth-matched off-route sites; negative values denote attenuation. Site medians are averaged within cells; symbols and whiskers are cohort means and 95% whole-cell bootstrap intervals (20,000 draws). **E**, Electrical calibration in the initial eight cells and a disjoint 45-cell cohort from the same mouse. The horizontal axis is the cohort-median axial/leak ratio at fixed axial resistivity $R_a = 150 \Omega \text{ cm}$; numerical labels give membrane resistance R_m in $\Omega \text{ cm}^2$. Shunt-minus-current contrasts are small and opposite-signed at $R_m = 15,000$, and increase at lower resistance. Symbols and whiskers are cohort means and 95% cell-bootstrap intervals after averaging sites within cell. These state-matched perturbations measure modeled transport, not plasticity or learning benefit.

dataset and thirteen eligible scans in the scan-complete analysis. These recordings sampled stimulus responses rather than learning.

Partners with more shared dendritic ancestry did not have reliably more similar condition-mean responses after controlling Euclidean separation and soma-to-contact path-length differences. Mean partial rank correlations were -0.069 for selected scans and -0.048 for all eligible scans; both target-bootstrap intervals included zero (Fig. 9A). This provides no evidence of preferential ancestry alignment in the sampled partners, at a different scale from population-level like-to-like wiring [44].

To establish the sensitivity of this small sample, response-level simulations preserved observed partners, shared inputs, repeated stimuli and seven-target inference, with noise calibrated to measured repeat reliabilities. Under the specified Gaussian ancestry-signal model, 80% detection required a mean simulated target-level partial rank correlation of approximately 0.25 (Fig. 9B). The observed repeat reliabilities used for attenuation vary across recordings (Fig. 9C). This conditional threshold leaves weaker effects, different signal geometries and unobserved partners unresolved (Methods; Supplementary Section S9).

A separate passive-tree model predicted each target's responses from its recorded partners. Its selected ancestry routes reached only 48.0% of mapped inputs, while one unrestricted fixed transfer profile already reconstructed 98.4% of exact credit and 97.6% of exact updates. Restricting that profile to incomplete route supports therefore tests transfer geometry more than example-dependent credit. Mean held-out NMSE was 0.788 for exact learning, 0.803 for ridge regression and 0.832 for the ancestry rule; the ancestry-minus-ridge difference was uncertain (0.029; 95% target-bootstrap interval, -0.013 to 0.074). Allowing oracle coefficients within the same ancestry span barely improved update reconstruction. Support, prediction, input-removal and reconstruction controls are grouped in SI (Supplementary Figs. S31–S32). This model diagnostic and the measured ancestry-similarity test constrain different parts of the anatomical proposal.

Discussion

Learning benefits from spatial selection when a common signal mixes incompatible synaptic changes. Dendritic morphology supplies candidate delivery patterns, teaching mechanisms supply their coefficients, and conductance regulates their gains. In the single-layer image classifiers, preserving neuronal identity supplies most of the benefit, including with random between-neuron feedback. Contextual conflict favors branch selection, supplied through feedback or local eligibility gating. Hierarchical distractors favor ancestry at a bandwidth predicted by the generator's coefficients. These results distinguish useful information from its physical carrier.

The interaction comparison explains why input rank alone is insufficient: pairwise and quartic targets share that spectrum and a compatible tree, yet learn different credit fields with different tolerance of fixed profiles. Spatial direction matters alongside dimension: even a nearly one-dimensional field can be poorly captured by uniform broadcast. These learned-state properties do not rescue the failed initialization selector. Intermediate dictionaries, adaptive profiles and alternative dynamics remain plausible solutions.

Conductance trees connect this distinction to computation: simple credit learns compatible monotonic teachers, whereas opposed tuning under contextual inhibition benefits from selection. A local inhibitory gate supplies that selection without oracle path coefficients, provided proximal credit remains ungated. The task supplies context and somatic error. Forward compatibility is a separate constraint: the scalar multi-affine cut criterion connects input grouping to hierarchical tensor descriptions and supports structure estimation from noisy examples [45–47]. Serial composition likewise helps remove distributed nuisance factors, but the LocalCA accuracy advantage of exact credit changes sign between the 180- and 600-epoch budgets while exact credit retains lower cross-entropy. The gap is still moving at the cap; forward depth and backward resolution need separate interpretation.

Obtaining coefficients remains a limitation beyond this explicit context gate. The separate encoder experiment exposes two obstacles: identifying a route from noisy cues and preventing coefficient leakage into incompatible routes even when identification is correct (Fig. 3D). Exploratory hard selection resolves the latter under noise-free calibration, but leaves a substantial noisy-cue deficit. Exact paths and projected coefficients elsewhere remain information-access controls. An adequate span alone does not ensure useful delivery.

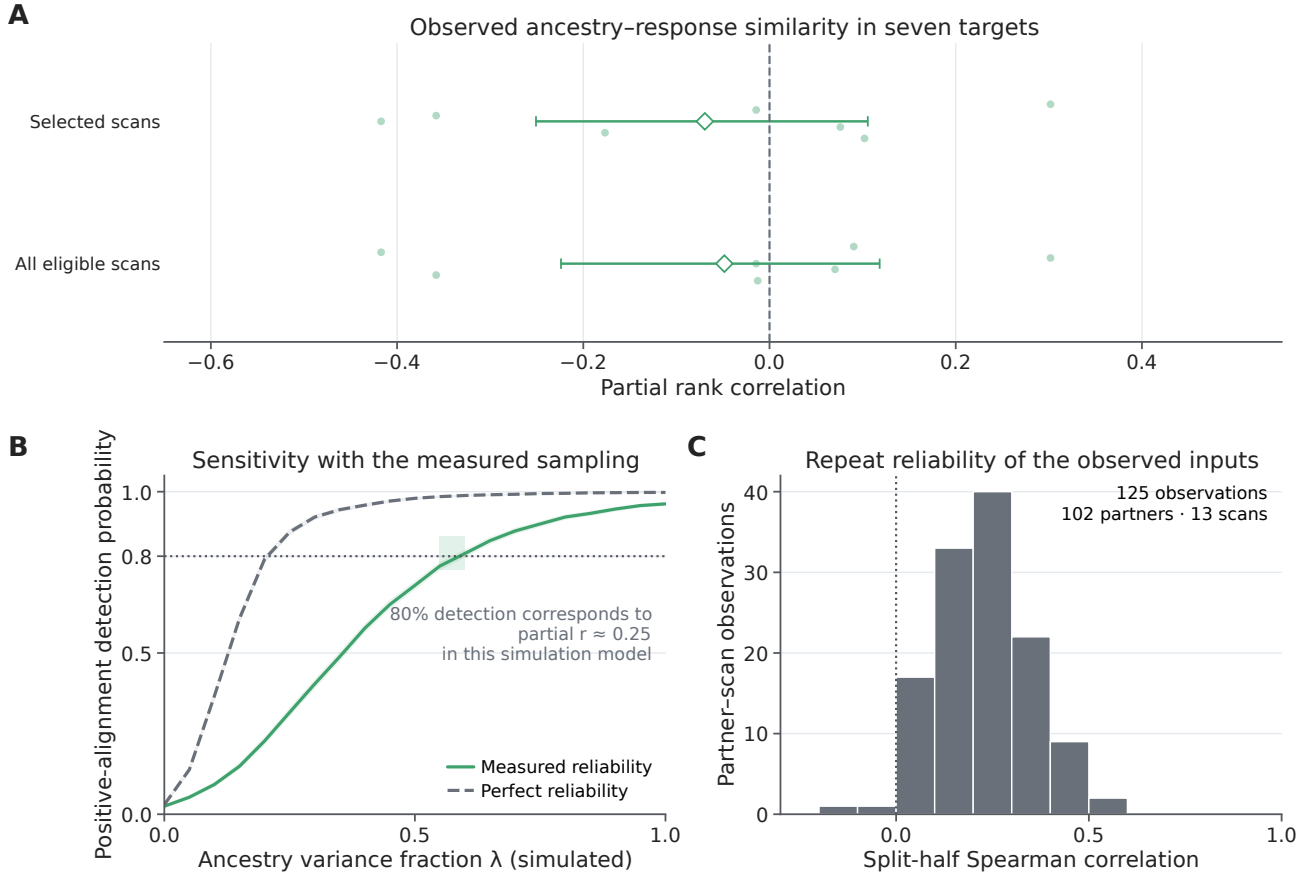


Figure 9: Measured ancestry alignment and the sensitivity of the available sample. **A**, Ancestry–response partial rank correlations after controlling Euclidean separation and contact-depth differences. Points show seven targets; diamonds and intervals show means and 95% target-bootstrap intervals. Selected scans contain 69 partners; all eligible scans comprise thirteen scans of the same targets. **B**, Conditional positive-alignment detection probability under measured or perfect response reliability. The simulated fraction λ allocates latent response variance to a Gaussian ancestry kernel; it is not an observed correlation. Each point summarizes 4,000 datasets analyzed by the exact two-sided seven-target signed-rank test at $P \leq 0.05$, also requiring a positive mean. Shading shows 95% Monte Carlo intervals. With measured reliability, the 80% threshold lies between $\lambda = 0.55$ and 0.60 , corresponding to mean partial correlation approximately 0.25 under this model. **C**, Measured split-half response correlations: 125 partner–scan records, 102 unique partners and thirteen scans. Negative values are retained here and clipped to zero only when calibrating nonnegative signal variance in the simulation. Support geometry, ridge comparisons and update reconstruction are reported together in SI.

Reconstructed arbors compress their modeled passive response fields beyond a common broadcast. Degree–depth surrogates explain much of this capacity, and the target fields themselves favor ancestry through passive transport under a somatic loss. Testing alignment beyond that construction requires measured fields or a model with different error sources or active dynamics; changing the passive perturbation family alone would not suffice. Focal shunting supplies a biophysical link by reweighting the baseline-weighted ancestry partition. Driving-force changes partly cancel adjoint changes, while cable state sets the spatial selectivity of the resulting gradient. This concerns sensitivity transport, distinct from inhibition that computes errors or regulates forward gain and load [48–51].

The small recorded cohort provided no evidence of preferential ancestry alignment; the offline learning diagnostic chiefly tests an almost fixed transfer profile. A direct experiment would compare plasticity below a focal inhibitory site with electrotonically matched off-route sites while matching somatic voltage, first-order current and outcome signal. Changing conductance state would test the predicted separation between shared and ancestry-selective effects [22, 52, 53].

The framework complements somatic prediction, segregated teaching, interneuron-mediated errors and burst plasticity [32, 34–36], alongside cable-learning and engineered gating models [39, 54–57]. Spikes, temporal eligibility, recurrence and channel kinetics may change the available profiles and coefficient estimation [58, 59]. The organizing question remains which task-relevant distinctions the arbor and learning mechanism can jointly preserve.

Methods

The Supplementary Information’s contents and cohort index (Table S11) locate complete derivations and protocols by scientific question. Resolved configurations, seeds, checkpoints, exclusions and source hashes accompany the numerical tables. Diagnostics reusing a cohort retain its original independent units.

Conductance models and delivery coordinates

In a directed tree, $[b_1, \dots, b_D]$ lists branching factors from soma to tips, giving $\sum_{d=1}^D \prod_{r=1}^d b_r$ nonsomatic compartments. A child c supplies output $a_c = f_c(V_c)$. With leak g_n^L , aggregate open excitatory/inhibitory conductances $G_n^X = \sum_{i \in \mathcal{I}_n^X} g_i^X x_i^X$ for contact class X , and child couplings $g_{c \rightarrow n}^{\text{den}}$, the steady voltage is

$$V_n = R_n^{\text{tot}} \left(g_n^L E_L + G_n^E E_E + G_n^I E_I + \sum_c g_{c \rightarrow n}^{\text{den}} a_c \right), \quad (R_n^{\text{tot}})^{-1} = g_n^L + G_n^E + G_n^I + \sum_c g_{c \rightarrow n}^{\text{den}}. \quad (9)$$

Here $x_i \geq 0$ is contact activity, $g_i \geq 0$ its conductance, and E_X its reversal potential. A terminal-to-soma pass evaluates the directed steady state. Artificial conductance trees use dimensionless leak one and reversals $E_L = E_I = 0$, $E_E = 1$. Monotone parameter transforms enforce nonnegative conductances. Branch-output nonlinearities, termed reactivations in the implementation, are identity functions when disabled. Both excitatory and inhibitory contacts learn in the regular-tree experiments.

For a reciprocal cable residual $\mathbf{F}(\mathbf{V}, \mathbf{g}, \mathbf{x}) = 0$, the nonsingular voltage Jacobian J_V gives the adjoint

$$J_V^T \mathbf{q} = \nabla_{\mathbf{V}} \mathcal{L}, \quad \partial \mathcal{L} / \partial g_i = -\mathbf{q}^T \partial \mathbf{F} / \partial g_i. \quad (10)$$

With passive current balance, the synaptic gradient is $x_i(E_i^{\text{rev}} - V_n)q_n$. This established sensitivity calculation supplies the reference for restricted credit [9, 10, 33, 39]. For output $a_n = f_n(V_n)$, activation error $\delta_n^a = \partial \mathcal{L} / \partial a_n$ gives voltage error $\delta_n^V = f'_n(V_n) \delta_n^a$. Image LocalCA replaces δ_n^a while retaining f'_n and exact local gate derivatives in eligibility. Fixed activation-space profiles can therefore vary in voltage space; exact supplied fields recover exact selected gradients.

The raw-additive control retains the graph but sums signed inputs without the conductance denominator or reversal-potential driving force. Historical normalized-additive controls additionally standardize voltage across stage coordinates within each example; Supplementary Section S2 defines those variants. The optional four- and five-factor rules add stage-level preconditioners and are described in Supplementary Section S1.

In $\mathbf{A}\Gamma$, nonzero diagonal gains preserve the dictionary span and oracle capture while changing coefficients and conditioning. This describes a passive shunt in its baseline-weighted ancestry partition, not arbitrary overlapping dictionaries. Stage count, feedback rank, external error dimension and state-dependent path resolution remain distinct (Supplementary Tables S1–S2).

Image feedback ladders and checkpoint diagnostics

The trained decoder supplies somatic activation errors $\delta_{i,u} = \partial \mathcal{L} / \partial a_{i,0,u}$ for example i and neuron u . Strict scalar feedback uses their mean, or their largest-magnitude signed coordinate when the mean’s magnitude is at most 10^{-6} , throughout the core including somatic gates. Neuron-specific feedback preserves each coordinate; exact transport additionally applies local path derivatives. The historical matched-width fallback retained neuronal identity at the 128-wide somatic stage, affecting 640 of 2,413,312 core parameters. A separately trained strict-scalar control tests that exception (Supplementary Section S2 and Source Data indexed in Table S12).

The new MNIST ladder uses 128-soma $[3, 3]$ shunting and raw-additive networks, with 40 excitatory and 20 inhibitory contacts per nonsomatic compartment. Within each architecture, rules share initial contact construction, initialization policy, decoder and data streams. Pixels are divided by 255 and flattened without further normalization or augmentation. Each seed divides the 60,000 training images into 48,000 training and 12,000 validation examples; all 10,000 official test images remain held out. Training uses 180 epochs, batches of 256 and the minimum-validation-cross-entropy checkpoint, without early stopping. Adam uses $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$ and no weight decay. Original rates are 0.0015 for input synapses and decoder, 0.0007 for branch couplings and 0.0005 for reactivation gates.

The projected $K = 1, 3$ rules average exact activation errors over the supports in Fig. 1B, preserving the field mean but not necessarily its norm. Both retain exact somatic errors and local derivatives. Decoder-only learning freezes the initialized core, verified by its hash. Three development seeds compare rate multipliers 0.3, 1 and 3 for every rule and architecture, giving 108 fits. Validation alone selects rates. Ten fresh paired seeds per architecture and rule give 120 selected-rate and 120 original-common-rate outcomes from 190 unique fits; 50 fits serve both policies. The projected- $K = 1$ addendum was frozen before fresh outcomes. Complete protocols, clipping and excluded implementation checks are in Supplementary Section S2.

Supplementary Section S2 gives the original fifteen-seed MNIST ladder, ten-seed Fashion-MNIST and twenty-seed raw-additive CIFAR-10 protocols; CIFAR uses a 20-soma $[3, 3, 3, 3]$ tree. Original gradient diagnostics appear in Fig. S7E–G. The fifteen-seed direct-feedback-alignment factorial replaces the decoder derivative by a fixed Gaussian matrix scaled by inverse square root of output dimension, paired with exact-readout references. Recipient permutations preserve coordinate values and bandwidth.

Common-checkpoint diagnostics compare rules at identical weights and states. Exact path transport is checked against automatic differentiation. New capture measurements use 2,048 held-out images at initial and exact-trained states; activation- and voltage-space metrics are kept separate. Gradient cosine, error magnitude by depth and mean-subtracted path energy are distinct summaries; none alone certifies learning performance. Historical depth and noise cohorts, including their task-identity and conductance-validity records, are specified in Supplementary Section S2.

Contextual conflict and ancestry routing

The branch-conflict task uses Eqs. 5–6, with independent opposite-class replacements of nonselected Fashion-MNIST inputs at probability χ . Twenty paired seeds cross $B = 2, 4, 8$ with eight doses. Gated-point and BP calculations receive equivalent updates; cyclic derangement changes recipients at fixed rank and support.

Under centered, symmetric class evidence, shared-mode compatibility is $s(\chi) = 1 - 2\chi(B - 1)/B$, with zero at $\chi_c = B/[2(B - 1)]$. Trained crossings are evaluated separately by mean-curve interpolation and each seed’s first sampled at-or-below-chance dose, retaining non-crossings. Checkpoint re-execution reproduces endpoints and compares exact/shared fields at both common and own trained states (Supplementary Fig. S9C–E).

The ancestry factorial contains eight eight-dimensional blocks. Context c_i and binary label sign s_i are balanced,

and inputs are $\mathbf{x}_{i,b} = a(b, c_i) s_i \mathbf{t}_b + \boldsymbol{\varepsilon}_{i,b}$, with seed-specific unit teachers $\mathbf{t}_b$ and Gaussian noise $\boldsymbol{\varepsilon}_{i,b} \sim \mathcal{N}(0, 1.1^2 I_8)$. The selected coefficient is one; sibling, same-half nonsibling and opposite-half coefficients are $-0.15, -0.45, -0.75$. A leaf permutation changes task-to-route alignment without changing the streams. For budget K , ancestry feedback has context-by-block entries $\Phi_{cb}^{(K)} = \sqrt{K/8} \mathbf{1}[g_K(\pi(c)) = g_K(b)]$, where g_K indexes contiguous groups and π maps streams to leaves. Thus every row has unit Euclidean norm.

The update is $\Delta \mathbf{w}_b = -(8\eta/N) \sum_i \delta_i \Phi_{c,b}^{(K)} \mathbf{x}_{i,\pi^{-1}(b)}$, with N training examples and rate η . Each seed supplies 2,048 training, 2,048 test and 1,024 switch examples. Full-batch descent uses rate 0.3, eighty training epochs and twenty switch epochs. Controls include deranged ancestry, depth-interleaved and random-sparse supports, dense rank- K fields and a training-informed PCA reference. The best matched control is the per-seed post-training maximum over the four non-PCA controls. Integer test counts preserve ties in the named Wilcoxon comparison; Holm correction covers four budgets. Alternative mean-sign-flip and historical BH analyses remain explicitly labeled in the Source Data indexed in Supplementary Table S12. The separate noisy-context encoder and cue-delay experiments are fully specified in Supplementary Section S3.

Update utility and prospective selection

Let the exact gradient be $\boldsymbol{\mu}$ and the delivered update estimator have mean $\mathbf{m}$ and covariance Σ_{route} at the current weights. For an L -smooth loss and a nonnegative step η , expected loss after the update is bounded above by $\mathcal{L} - \eta \boldsymbol{\mu}^\top \mathbf{m} + (L\eta^2/2)(\|\mathbf{m}\|^2 + \text{tr} \Sigma_{\text{route}})$. Optimizing this expression gives utility $U = [\boldsymbol{\mu}^\top \mathbf{m}]_+^2 / [2L(\|\mathbf{m}\|^2 + \text{tr} \Sigma_{\text{route}})]$, with zero for a zero denominator. Utility certifies one-step progress under the stated moments; it is not a long-horizon ordering.

Synthetic quadratic screens use sixteen Haar coordinates and a designed signal/noise crossover, without a forward neuron (Supplementary Fig. S2). Retrospective factorial utility uses actual full-batch updates and a global logistic-curvature bound; maximum bandwidth matches its selection success. Supplementary Figs. S33–S34 retain the failed 12,800-fit prospective selector, its loss to fixed maximum-budget and rank-only baselines in every seed, and the 25,600-fit dynamics extension with evolving noise, observed-context-count controls, sealed choices and computational costs.

Interaction compatibility, fixed profiles and learned spectra

For a subtree leaf set S , C_S arranges the target’s Fourier interaction coefficients by products inside and outside S ; C_S° removes the constant inside row. Scalar multi-affine trees represent the target exactly if and only if every centered cut has rank at most one. If $\mathcal{S}(T)$ contains all proper cuts of tree T and singular values are ordered from largest to smallest,

$$\text{MSE}(f, \hat{f}_T) \geq B(T; f) := \max_{S \in \mathcal{S}(T)} \sum_{j \geq 2} \sigma_j(C_S^\circ)^2. \quad (11)$$

Overlapping cut tails cannot be added as a lower bound. Supplementary Section S4 gives the constructive proof, minimum-depth algorithm and its relation to hierarchical tensor representations. The twelve-tree census, noisy-query estimator, end-to-end learning pipeline, conductance grouping controls and Boolean truth tables remain in Supplementary Figs. S11, S13–S14, S17 and S35. The theorem’s assumptions do not extend automatically to general conductance ancestors.

Three development seeds select rates with equal budgets; twenty fresh seed blocks pair matching and quartic conditions across trees, assignments, initialization, inputs, additive noise and minibatch order. Exact, unit, calibrated and sign-only profiles share local eligibilities; calibration averages path derivatives over 256 unlabeled initialization examples. Selected/common rates under SGD and Adam yield 432 development and 1,200 fresh fits. Coefficients stay within $[-2, 2]$, with bound contacts recorded. A balanced extension replays all seven original checkpoints through 1,024 steps, then continues all 720 unique selected/common-rate algebraic trajectories to 16,384. Terminal and validation-selected extensions describe the same seeds (Supplementary Section S4). Noisy held-out NMSE divides by clean target variance: one for matching and nested, one-half for quartic. All have noise variance 0.0225. The primary Adam contrast subtracts the pairwise calibrated-minus-exact gap from its quartic counterpart.

For evaluation-example-by-site path matrix Q , the best fixed profile is the leading right singular vector. Its capture is $\sigma_1(Q)^2 / \sum_j \sigma_j(Q)^2$, an oracle reconstruction with example-dependent amplitudes. Additional diagnostics evaluate uniform, calibrated and ancestry profiles, loss-residual weighting, eligibility-weighted updates and root-mean-square normalization of each site. Participation-ratio effective rank is $(\text{tr}C)^2 / \text{tr}(C^2)$ for uncentered $C = Q^\top Q / N$. At fixed weights and inputs, changing labels leaves the path field unchanged; eighty relabeling controls verify this. Independent checkpoint analysis and faithful replay of 600 earlier fits provide numerical and trajectory checks. Complete definitions, pairing audits, optimizer effects and the positive-conductance boundary are in Supplementary Sections S4–S5.

Conductance trees with context-dependent feature tuning

The seven-compartment model uses identity branch outputs in Eq. 9. Two independent $z_j \sim \mathcal{U}[-2, 2]$ generate positive inputs $\exp(\pm z_j)$ for both subtrees. Every terminal has two excitatory and two inhibitory gains; binary context inhibits one proximal compartment. All 24 conductances are positive and parameterized by logarithms. Students start from terminal gains one, proximal inhibitory gains 15, daughter couplings 20 and 0.2 per subtree, and soma couplings four. These proximal and coupling values match the teacher’s nominal scaffold, not its realized weights. Independent Gaussian log perturbations have SD 0.4 in students and 0.15 in teachers. Full generators and eligibility derivatives are given in Supplementary Section S5.

The local-gate follow-up used twenty new paired seeds and nine rules at Adam rates 0.01, 0.03 and 0.1, giving 1,080 trajectories. Each task supplied 2,048 training, 1,024 validation, 4,096 test and 512 unlabeled calibration examples. The primary rate 0.03 was fixed from earlier work, not selected on these seeds. Minibatches contained 128 examples; logarithmic conductances stayed within $[-7, 7]$. Minimum validation NMSE selected states within fixed 4,096- and 16,384-update windows. The somatic error was $(V_s - y) / \text{Var}_{\text{train}}(y)$; NMSE uses clean target variance. At a terminal with parent p , the hard gate multiplied this error by $\mathbf{1}[a_p = 0]$, where proximal inhibitory activity a_p was zero or ten. Proximal and somatic multipliers remained one. The continuous gate used $1/(1 + g_p^I a_p)$. Both combined these signals with the same exact local eligibilities and accessed no exact path derivatives.

Controls used exact credit, unit and initially calibrated broadcast, two weighted distal oracle profiles with unit proximal credit, three oracle profiles including a proximal column, reversed distal gating, and gating at proximal sites as well. Two primary paired sign-flip tests—opposed broadcast-minus-gate error and its opposed-minus-aligned difference—were Holm-adjusted together. Other comparisons and 95% whole-seed bootstrap intervals were descriptive. The protocol followed exploratory results but preceded execution on new seeds. Supplementary Section S5 and Figs. S18–S20 give full rules, cancellation diagnostics, development, wider-bound and local-gate controls.

Serial depth, sensor alignment and optimization

Physical-depth networks contain 64 somata and eight nonsomatic modules each. D1–D4 reallocate these modules across branching vectors $[8]$, $[2, 3]$, $[2, 1, 2]$ and $[1, 1, 2, 2]$ in the three- and four-tier comparisons; the two-tier D2 tree uses $[4, 1]$. Contacts and resources are matched; serial composition changes. Grouped-point controls connect every module directly to the soma. Raw-additive controls retain the graph without division, while flexible MLP controls are approximately parameter matched.

Excitatory blocks carry class evidence multiplied by lognormal nuisance factors. Inhibitory blocks sense those factors with alignment α : zero removes trial-specific information while preserving the sensor distribution, and one reports the factor. The generator floors noisy input rates before multiplying them by gains. Nested, flat and local-ratio families distinguish distributed hierarchical support from generic factor cancellation and locally available ratios. Full generators, floors, sensor-placement controls and contact inventories are defined in Supplementary Section S6.

Original cohorts use ten paired seeds, a 180-epoch cap, thirty-epoch patience and validation-based state selection. BP and LocalCA differ in rates and parameter groups; exact-path LocalCA still computes exact selected gradients. Source checks of one hundred original trajectories verified configurations and final metrics: all fifty D3 fits across five credit/optimizer arms reached the cap with falling validation loss. Sixty source-consistent restarts extend these

661 fifty D3 fits and ten exact-BP D1 references to 600 epochs, selecting minimum-validation-loss states within each
662 budget. All D3 fits and two D1 references reach the larger cap; eight D1 references stop early. D3 selected states lie
663 near the cap while validation loss continues falling. Supplementary Section S6 retains complete histories, paired
664 accuracy/loss contrasts and source-concordance discrepancies.

665 **Anatomical dictionaries and common-mode controls**

666 Observed MICrONS skeletons and mapped contacts define soma-rooted branch-to-branch segments. Initial eight
667 and disjoint 47-cell cohorts come from the same mouse; Pinky is a second mouse, with ten of twelve selected cells
668 meeting the inherited typed-contact criterion. Each subtree route contains the input-bearing descendants of its origin.
669 Contact labels are direct where available and otherwise proxy-based as specified in Supplementary Section S7.
670 Compression preserves path length but only approximates resistance; direct-label, mapping and geometry sensitivities
671 are retained.

672 The modeled passive perturbation field consists of reciprocal passive-cable gradient changes induced by small
673 focal shunts. Every corrected dictionary receives one constant column and $K - 1$ spatial columns. Ancestry selection
674 uses inhibitory fraction times excitatory-area-weighted descendant fraction without field outcomes. Depth bins
675 are reparameterized with the same constant while preserving their span; random routes, independent site shuffles
676 and degree–depth surrogates provide controls. The weighted common component is removed before computing
677 residual-energy capture. A common-constrained SVD oracle supplies the rank-budget ceiling. Numerical rank,
678 coverage and nonzero density are reported separately; equal column count alone does not match those resources.

679 The initial eight cells provided post hoc development for common-mode correction; the protocol preceded
680 evaluation of new v661 and Pinky outcomes. Two hundred randomized controls are averaged within each cell. At
681 eight columns, all 47 v661 cells and eight Pinky cells qualify; other eligible budgets are retained without substituting
682 for missing eight-column outcomes. Paired cell-bootstrap intervals and Holm-adjusted comparisons describe within-
683 cohort consistency, not animal-population replication. The older route-generated positive controls use separate fields
684 and endpoint definitions (Supplementary Section S7).

685 **Focal shunting and the electrotonic boundary**

686 Up to sixteen focal sites per cell span depth quartiles and require at least three excitatory descendants and three
687 comparison sites. Normalized E/I scales are 0.35 with reversals one and -0.2 . Shunt doses are 0.25, 0.5, one and
688 two times local membrane-plus-synaptic conductance. A matched injection supplies first-order current $\kappa_k(E_I - V_k)$
689 without changing the conductance matrix. A separately solved somatic current restores baseline voltage and is held
690 fixed during differentiation. The squared-loss target is one normalized unit below baseline; finite differences check
691 the adjoint gradients.

692 Localization subtracts depth-matched off-route from descendant medians of $|\log[(|\gamma'_i| + \varepsilon)/(|\gamma_i| + \varepsilon)]|$, with
693 $\varepsilon = 10^{-15} \max(1, \max_i |\gamma_i|)$. Because this is unsigned, signed gradient changes are reported separately. Adjoint and
694 driving-force substitutions in $\gamma_i = q_i(E_E - V_i)$ distinguish transport from local voltage effects. Sites and relation-
695 reassignment templates are nested within cells. The passive ancestry-partition identity and its dictionary qualification
696 are derived in Supplementary Section S8.

697 Physical calibration uses cylindrical axial conductance $\pi r^2/(R_a \ell)$ and leak $2\pi r \ell/R_m$, expressed in nS, with
698 spherical soma area $4\pi r^2$. Here r and ℓ are segment radius and length, R_a axial resistivity and R_m specific membrane
699 resistance. Sweeps include $R_a = 100, 150, 250 \Omega \text{cm}$ and $R_m = 300\text{--}30,000 \Omega \text{cm}^2$, fixed and locally normalized
700 doses, and background conductance. The disjoint calibration cohort contains 235 eligible sites in 45 cells. Geometric
701 fallbacks and dose definitions are specified in Supplementary Section S8.

702 The weak-channel check uses $R_a = 150$, $R_m = 1,000$ in those units and background conductance equal to leak.
703 Median maximal Na, K, Ca and HCN conductances are 0.06, 0.10, 0.025 and 0.05 times leak; NMDA is 0.15 times
704 excitatory conductance. Sixty-four draws per cell passed the predefined steady-state checks. Responses use rank-one
705 updates of the fixed baseline Jacobian, including voltage-gate derivatives, followed by somatic state matching. They
706 are local linearizations rather than newly solved nonlinear equilibria at each dose (Supplementary Fig. S29D,E).

707 **Measured-response prediction and geometric diagnostics**

708 The seven-target cohort includes thirteen eligible scans. Trial responses are the mean of the supplied ROI fluorescence
709 traces in `RoiResponseSeries` over each stimulus interval; our analysis did not deconvolve the traces or apply a
710 reliability correction. Repeated responses are averaged by stimulus condition for the ancestry-similarity analysis;
711 partial rank correlations control Euclidean separation and soma-to-contact path-length differences. Offline learning
712 holds out stimulus identities. Training-only percentile transforms produce nonnegative presynaptic proxies, and each
713 partner enters at its dominant mapped contact segment. A linear somatic readout predicts the standardized target
714 response. Only input-conductance credit differs across rules; readout and bias gradients remain exact.

715 Four-route dictionaries are selected without target responses and receive the projected baseline transfer profile
716 multiplied by the scalar readout error. Supplementary Section S9 defines delivery and reconstruction. Fig. S31
717 reconstructs supports from saved segment identifiers, illustrating the median-input-count scan with identifier-based
718 tie breaking; all scan supports are available. At common exact-trained states, field and eligibility-weighted update
719 reconstruction compare the fixed profile, per-trial field projection and the best within-span update. These are oracle
720 diagnostics rather than newly trained rules. Reproducibility comparisons report endpoint discrepancies across 130
721 exact-rule re-executions.

722 Ridge regularization uses training data within each outer split and the same measured predictors. Input removal,
723 predictor noise and reliability masks are separate sensitivities; none uses held-out information for selection. Ten
724 split/initialization replicates are averaged within scans and then within seven targets. Imposed-alignment quadratic
725 fields and the six-animal external signed-credit reanalysis are separate datasets and tests, described in Supplementary
726 Sections S7 and S9, respectively.

727 **Statistics and computational provenance**

728 Confirmatory choices precede outcome inspection but are not described as public preregistration. Pilots, post hoc
729 analyses and same-seed reruns are identified separately. Effect sizes accompany experiment-specific 95% intervals,
730 or adjusted 97.5% intervals where prespecified. Tests are two sided unless a directional protocol states otherwise.
731 Named multiplicity families and exact tie treatment are retained in Source Data; null results imply equivalence only
732 when a margin is tested.

733 For the sensitivity simulation, a positive-semidefinite shared-path covariance generated joint tuning across the
734 102 observed presynaptic roots. We mixed ancestry-correlated and independent reliable tuning, then added noise
735 calibrated to each recording's split-half reliability and actual repeat counts. Four thousand paired datasets at each
736 of 21 signal fractions were analyzed with the original partial-rank statistic and an exact seven-target signed-rank
737 test. Detection required two-sided $P \leq 0.05$ and a positive mean effect; Wilson intervals describe Monte Carlo
738 uncertainty. The model and analysis were frozen before simulated outcomes. Supplementary Section S9 gives the
739 covariance, reliability calibration, dependence assumptions and exact test resolution; the released pair-level inputs
740 permit reproduction.

741 Independent units are paired training seeds, reconstructed cells, postsynaptic targets or external animals. Sites,
742 contacts, scans, splits, checkpoints and randomized draws remain nested. Seed counts reflect the experiment's
743 purpose; no prospective parametric power calculation was performed. Source hashes certify retained inputs and
744 implementations, not unavailable historical runtime states. The matched credit and new conductance studies recorded
745 Python 3.10.13, NumPy 2.2.6 and pandas 2.3.3. The new MNIST controls used PyTorch 2.9.1/CUDA 12.8 and
746 torchvision 0.24.1 with NumPy 2.2.6; the depth extension used Python 3.10 and PyTorch 2.9.1. The isolated reviewer
747 environment was validated with NumPy 1.26.4, SciPy 1.15.3, PyTorch 2.9.1, torchvision 0.24.1 and Matplotlib
748 3.10.9. Cohort-specific runtime records distinguish executed experiments from this installation check. Supplementary
749 Section S10 consolidates statistical families and release records; the question-specific sections retain their exclusions
750 and numerical checks.

751 OpenAI Codex assisted with code, statistical and consistency audits, figure preparation, literature checks and
752 language editing. The authors remain responsible for design, interpretation, numerical accuracy, citations and the
753 final manuscript.

Ethics and data reuse

No new experiments involving animals or human participants were performed. All biological analyses reused public MICrONS, DANDI, and published animal source data. The source studies describe the original data collection and institutional approvals [22, 42, 60].

Data availability

MICrONS anatomy and connectivity are available through the public CAVE datastack minnie65_public. The primary cohort used materialization 1822, skeleton-service reconstructions, and the synapses_pni_2 table. The disjoint sensitivity cohort used the public v661 static release. Secondary target-structure labels used only in the full-contact analysis came from the in-development synapse_target_predictions_ssa_v2 table. These labels are distributed only as derived data and remain subject to the source-data access terms. All principal structural findings were repeated without them.

The independent-animal replication used the MICrONS phase-1 layer-2/3 Pinky v185 fixed meshes, soma-valence table, and dense synapse table archived at Zenodo DOI 10.5281/zenodo.3710459 [61]. File sizes, Zenodo MD5 checksums, and local SHA-256 checksums are recorded in the supplied manifest. Volumetric imagery is archived in BossDB under DOI 10.60533/BOSS-2021-T0SY, and co-registered functional data are available as DANDI:000402. The supplied manifest records the exact asset and table identifiers, and the source publication describes the full data release [42].

Public benchmark datasets are available from their original repositories. The inhibitory-census tables and SWCs are available with the source record for Schneider-Mizell et al. [60]. The Francioni et al. workbook is available through the Source Data link associated with the Nature article [22]; its checksum and the sheet names used here are recorded in the manifest. Derived cell and exclusion manifests, numerical source data underlying every figure, and non-restricted summaries are supplied with this manuscript. The measured-response release includes the analyzed partner-pair tables, contact geometry, recording identifiers and repeated-stimulus response arrays needed for the sensitivity analysis. Authentication tokens and externally hosted raw caches are not redistributed.

Code availability

The accompanying reviewer software archive contains the complete training implementation, exact-gradient and perturbation analyses, versioned configurations, tests, figure scripts, and byte-identified snapshots of the critical training components under the MIT License. A separate Source Data archive contains the numerical data underlying every display item. These archives are available to editors and reviewers with the submission. The same version will be released publicly under the MIT License in a versioned repository and permanent archival record upon publication.

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Author contributions

H.S. developed the computational framework, performed the analyses, and wrote the initial manuscript. M.R. contributed to the theory of local credit assignment, its implementation, and its interpretation. B.L.S. supervised the work and contributed to the biological framing, study design, and manuscript. All authors reviewed and revised the manuscript.

Competing interests

The authors declare no competing interests.